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Environmental regulation and productivity growth in the euro area: Testing the porter hypothesis

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ABSTRACT

This paper examines the impact of changes in the stringency of environmental regulations on productivity growth. We exploit several data sources, including the OECD Environmental Policy Stringency Index and balance sheet information from ORBIS and iBACH, to test the Porter hypothesis, according to which firms' productivity can benefit from more stringent environmental policies. We estimate the regulatory impact over a five-year horizon using panel local projections. To identify the direction of the effects, we estimate CO_2 equivalent emissions for all firms in our sample using a machine learning algorithm. As suggested by the country-level analysis and confirmed by the firm-level analysis, policy tightening negatively affects productivity growth of high-polluting firms and to a larger extent than that of their low-polluting peers. Hence, we do not find support for the Porter hypothesis in general. However, not all policies have the same impact – non-market based policies are the most detrimental to productivity growth – and not all highly polluting firms are affected in the same way – the negative impact is mitigated for large firms, which may benefit from easier access to finance and greater innovativeness.

1. Introduction

In establishing the objective of reaching 'net zero' greenhouse gas emissions by 2050 and launching the Green New Deal, the EU has set a clear direction for the future. In addition to potent EU-level policies to tackle climate change, such as the EU Emissions Trading Scheme (ETS), a growing number of national governments are signing up to their own greenhouse gas emission targets and adopting increasingly stringent policies to achieve their ambitions at the national level. The policies implemented run the entire gamut of command-and-control, market-based and technology support options, from mandatory standards to carbon pricing and R&D subsidies. These policy changes are likely to trigger a wide-ranging transformation of the EU economy and reallocation of resources across sectors. This is especially true when combined with stimulus packages adopted in the wake of the COVID-19 crisis, which also aim to support 'green' economic activities, albeit to varying degrees. The economic effects of climate regulation and environmental regulation more generally are therefore of key interest to policy-makers at both national and EU level.

Regulation is traditionally seen as a hindrance to economic activity, at least in the short to medium term, as it raises costs without increasing output and restricts the set of available production technologies and outputs. In contrast to this view, the Porter hypothesis (Porter and Van der Linde, 1995) suggests that, under certain conditions environmental policies can spur innovation and

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by doing so enhance productivity, which can offset or even outweigh the costs of the policy. As evident from the literature review, the number of existing empirical analyses regarding the validity of the Porter hypothesis is large but results remain inconclusive, with some studies suggesting it holds while others arguing the opposite. Even less is known about the specific conditions and firm characteristics under which it more likely to hold or the opposite. The majority of relevant published studies are focused on single countries or industries and suffer from limited external validity, while potential endogeneity issues are rarely addressed in a robust manner.

This paper probes the link between environmental policy stringency and productivity growth in six euro area countries (Germany, Belgium, France, Italy, Spain and Portugal) in the period between 2003 and 2019. More specifically, we assess the dynamic impact of environmental regulation on euro area firms' total factor productivity (TFP) and labour productivity (LP) growth and its aggregate consequences on competitiveness. The use of firm-level data is fundamental to analyse heterogeneous impacts of policies across firms differing in size, innovativeness or financial resources. These heterogeneous effects are often hidden in aggregate data. The granular analysis is complemented with country-level insights to gauge the aggregate impacts of policies. Lastly, we explore the differences between different types of policy (market-based vs non-market-based vs technology-support) in their impact on productivity growth. We find that high-polluting firms, which are more exposed to regulatory changes, are significantly negatively affected, so that we have to reject the strong Porter hypothesis for the whole population of firms. It is, however, important to distinguish between policy types as they affect TFP growth differently. In case of high-polluting firms, technology support policies only have transitory negative effects before boosting TFP growth; market-based policies have small but negative effects; and non-market based tools reduce TFP growth the most. Among high polluting firms, the impact of policy tightening varies: more resources, as proxied by a higher equity ratio and more experience with patenting can ameliorate the negative impact of stricter environmental policies.

This paper contributes to the empirical literature examining the Porter hypothesis in several ways. First, our panel data set spans multiple countries and sectors. By including firms from sectors beyond manufacturing and using ORBIS and iBACH data, we exploit the largest firm-level data set for the analysis of the Porter hypothesis to date, with data on more than three million individual firms. Second, our contribution relies on a novel identification strategy based on the assumption that environmental policy is likely to affect firms differently depending on their exposure to regulation, with firm-level emissions used as a proxy for this exposure. A similar approach has been applied by Albrizio et al. (2017) albeit assuming that the emission intensities of all firms within a sector (and across countries) are identical. This is, however, in stark contrast to empirical findings (see Lyubich et al., 2018) that show large emission heterogeneity even within narrow sectors. Scarcity of firm-level pollution data, especially for smaller firms, would normally hinder the deployment of this approach at firm-level. We overcame this problem by estimating emission equivalents using a Boosted Trees machine learning model based on firm-level emission and financial account data. Third, we contribute by analysing the dynamics over a five-year horizon and explore firm heterogeneities in detail. We use local projections to estimate the non-linear impulse responses of TFP growth to a change in environmental regulation, defined as a positive change in policy stringency. This is motivated by the fact that environmental policies might have different short and medium-term impacts. The local projection methodology allows us to describe dynamic effects of changes in regulation and to capture heterogeneous impacts across firms resulting from differences in financial and innovative abilities.

The paper is structured as follows: Section 2 reviews the literature and the importance to control for endogeneity. Section 3 describes the macro and firm-level data as well as the emission estimates, which are used in the local projection framework explained in Section 4. Section 5 presents the effects of different types of environmental policies on productivity growth at country and firm-level and elaborates on the role of financial sources and research experience. Section 6 concludes.

2. Literature review

The Porter hypothesis has a number of theoretical offshoots. The 'strong' version states that increases in environmental regulation stringency might raise overall productivity, while the 'weak' version holds that optimally designed environmental regulation spurs green innovation. The 'narrow' version predicts that market-based tools are more effective in boosting innovation than command-and-control policies. While we analyse the 'weak' version in a related paper (not published yet), the focus of this paper is on the evaluation of the strong (and narrow) version at country- and firm-level.

Various sources of endogeneity complicate the identification of a causal link between regulation and economic outcomes including productivity. A great concern is omitted variable bias, which can arise as a result of other regulations (e.g. labour or financial market), expenditure on R&D or trade that affects both TFP growth and decisions on environmental stringency. Moreover, Albrizio et al. (2017) point out that simultaneity issues may arise if, for instance, good environmental performance in given industries facilitates adoption of more stringent environmental policies or if firms that are performing poorly are able to successfully lobby against more stringent policies. The present section therefore reviews the most relevant literature with a particular focus on the identification strategy chosen by the authors.

Most prior studies examining the strong version of the Porter hypothesis do not sufficiently address potential endogeneity issues, which may result in biased estimates (Hille and Möbius, 2019; Brunel and Levinson, 2020). Cohen and Tubb (2018) also support this finding and qualify this in their meta-analysis by claiming that studies are more likely to yield evidence supporting the strong version of the Porter hypothesis when an instrumental variable is implemented for environmental regulation. Overall, the evidence presented is inconclusive both with regards to the significance and direction of the effect (Ambec et al., 2013; Kozluk and Zipperer, 2015; Cohen and Tubb, 2018).

The majority of studies exploring the relationship between environmental regulation and productivity focus on specific countries and industries and, in many cases, specific legislation. While studies of a single policy reform (e.g. Berman and Bui, 2001; Lutz,

2016; Marin et al., 2018; Löschel et al., 2019) tend to offer a more straightforward path to causal identification of the impact of a specific policy instrument on productivity, there is often a limited external validity. These studies provide little information about the global impact of environmental policy and the possible economy-wide costs and transition risk that it may or may not introduce. We believe that there is a value in contributing to the body of knowledge regarding the economic impact of environmental policies using firm-level data also through multi-country and multi-sector studies using composite policy indices such as the Environmental Policy Index. These insights are particularly relevant to EU-level policy-makers, who are interested in the economic impacts of environmental policy beyond specific instruments and regulated sectors. In terms of enhancing our understanding of transition risks and possible regulatory responses to it, these studies better reflect the real life context in which firms are affected, directly or indirectly, by an (often overlapping) mix of policies at any particular time.

More closely related to our enquiry are those contributions considering various aggregate measures of environmental policy stringency and which therefore cannot exploit inclusion criteria in specific policies. Many of these studies tend to avoid making strong claims about causality. Lanoie et al. (2008), for instance, analyse the relationship between the stringency of environmental regulation and total factor productivity (TFP) in the Quebec manufacturing sector. Their results suggest that the contemporaneous impact of environmental regulation on productivity is negative and turns positive in models using lagged measures of environmental policy stringency. Hamamoto (2006) finds that environmental command-and-control regulations led to an increase in innovation (proxied by R&D spending) and consequently to an increase in productivity growth in a sample of five Japanese manufacturing sectors over 20 years. He likewise refers to "positive association" rather than a clear causal relationship. More recently, a similar approach was taken by Lee and Lee (2022) who use a multilevel linear model and their results indicate that environmental protection expenditure has a negative correlation with TFP in Korea.

In terms of cross-country studies, Hille and Möbius (2019) is a notable recent study that tests the Porter hypothesis at industry level and tackles endogeneity concerns by using fixed effects and a dynamic panel generalized method of moments (GMM) estimator. Their positive regulatory effects on productivity becomes insignificant once simultaneity is controlled for.

The closest study to ours is that of Albrizio et al. (2017), which uses the Environmental Policy Stringency (EPS) composite index to measure the impact of environmental policy on productivity growth. The study combines insights from firm- and industry-level analysis and finds that the tightening of environmental policy has a positive short-term effect on productivity growth at the industry level in the manufacturing sector. At the firm-level, only a minority of the firms (the most productive ones) experience productivity gains following a tightening of environmental regulation. They also find that temporary effects tend to differ by policy type, with market-based instruments having a more robust positive effect on productivity growth than non-market-based measures.

Similar to ours, the identification strategy in Albrizio et al. (2017) is related to Rajan and Zingales (1998), who explored the role of financial development on economic growth using industry-specific financial dependence as the exposure variable. By now, this approach of testing the underlying mechanism is common in the analysis of impacts of national-level policies of all types on developments at the industry and firm-level. The identification strategy of Albrizio et al. (2017) relies on the fact that environmental policy is likely to affect industry productivity heterogeneously depending on the industry's exposure to the regulation. A higher environmental dependence (proxied by pollution intensity) increases the industries' exposure to the country-level environmental policies and hence the potential economic effects. Contrary to our identification strategy, they assume that the emission intensity of each firm in an industry is the same, which contradicts our data (see Fig. A.1) and the findings of Lyubich et al. (2018). Beside our firm-specific exposure estimates, there are several other relevant differences to our analysis. We expand the sample period by ten years which allows us to describe dynamic effects up to five years ahead and analyse more recent stringent policies. Due to the update of the EPS index (Kruse et al., 2022), we can distinguish between command-and-control policies and subsidies which were analysed together as non-market measures by Albrizio et al. (2017). As we expect the effects of stricter limits and higher subsidies to be different on productivity, this might obfuscate their results. Further, we generalize the results by including also other than the manufacturing sector and we make the sample more representative by adding more small firms with the iBACH data. Finally, we analyse various sources of heterogeneities among firms beside the role of the distance to the technology frontier.

3. Data

Our analysis combines a variety of data sources to describe the effects of changes in environmental regulation on productivity. These include aggregate indicators from the OECD and Eurostat to describe macroeconomic changes and environmental policies as well as granular firm and emission data from ORBIS, iBACH and Urgentem. Our data set spans the period from 2003 to 2019 and includes 15 EA countries for the aggregate analysis and firms from 6 EA countries for our firm-level analysis.[1]

3.1. Measuring environmental policy stringency

In order to capture the development of environmental regulation, we use the index of environmental policy stringency (EPS) developed by the OECD (Botta and Koźluk, 2014; Kruse et al., 2022). The current version of the EPS index covers 34 OECD countries over the period 1990–2020 and summarizes environmental policy stringency across selected instruments. The indicator consists of three components: a market-based, a non-market based and a technology support sub-indicator. The market-based component groups

[1] Austria, Belgium*, Estonia, Finland, France*, Germany*, Greece, Ireland, Italy*, Luxembourg, the Netherlands, Portugal*, Slovakia, Slovenia and Spain* - * indicates availability of firm-level data.

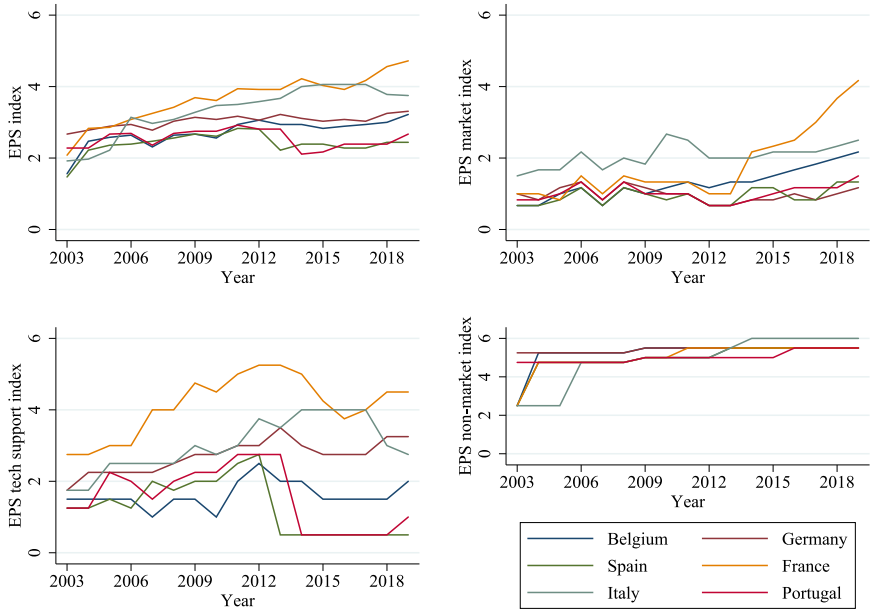

Fig. 1. EPS index.
Note: Development of the EPS index and its sub-indicators (market, non-market and technology support) between 2003 and 2019. Range between 0 and 6, 6 represents the most stringent policies among OECD countries since 1990.

instruments which assign an explicit price to externalities (taxes: CO_2, SO_X, NO_X, and diesel fuel; trading schemes: CO_2, renewable energy certificates and energy efficiency certificates). The non-market component clusters performance standards (emission limit values for SO_X, NO_X and PM, limits on sulphur content in diesel). Finally, technology-support policies capture green R&D subsidies (per GDP) and adaption support measures like feed-in-tariffs. All indicators range from 0 to 6 with higher numbers being associated with more stringent (i.e. more pollution mitigating) environmental policies. This implies that higher taxes, lower limits and more subsidies increase the indicator values.[2]

Fig. 1 provides an overview of the evolution of the EPS index as well as the three sub-indices in the countries relevant for the firm-level analysis. While we observe a trend towards more stringent policies, there is substantial heterogeneity across countries and sub-indicators. The gap between the least and the most stringent country in the overall index increased from 1.2 in 2003 to 2.4 in 2019. The non-market index indicates a comparatively strict environment in the six countries, while the market index shows a low level of stringency, with France being the only country with large changes. The technology support index stands out, as there are also meaningful reductions in stringency due to austerity measures during the European sovereign debt crisis. The variability of the indices after controlling for year and country fixed effects can be seen in the Appendix (Fig. A.2).

The intention of the EPS index is to proxy the environmental policy stringency of the whole economy, although several environmental regulations are sector specific. Botta and Koźluk (2014) focus on regulations of the energy (and transport) sector curbing greenhouse gas emissions and air pollution. These two sectors are present and important in all countries, face high pollution intensity and are regulated by well known instruments over a reasonable long time period. By capturing regulations of upstream activities that impact other sectors indirectly, Botta and Koźluk (2014) argue that the policy stringency measured by the EPS is a reliable proxy for the overall aim to reduce negative emission externalities. Note that thereby all firms (using energy) are affected by the regulations but they are differently exposed depending on their technology. Comparing the EPS indicator to other environmental policy stringency indices, like the economy-wide stringency indicator of World Economic Forums Executive Opinion Survey, supports reliability of the EPS.

Similarly to Albrizio et al. (2017) and De Santis et al. (2021), our study uses the change in the EPS indicator rather than the level. The reason is that new investments into abatement technology are more likely to be spurred by a substantial change in regulatory stringency. When focusing solely on a single country, the presence of a fixed level of environmental regulations, in terms of the difference in relative prices of inputs, will not by itself motivate firms to alter their production methods. Furthermore, EPS (sub-indicator) levels are partly non-stationary, while first differences are stationary, thus facilitating empirical analysis. Since the Porter hypothesis concerns the effects of more stringent policies, we specifically focus on positive changes of the indicator to

[2] Note that we do not only distinguish between sub-indicators to identify differences in effects, but also because more stringent market and non-market polices might harm firms, while more technology support is expected to be beneficial and might bias the overall indicator's impact towards zero.

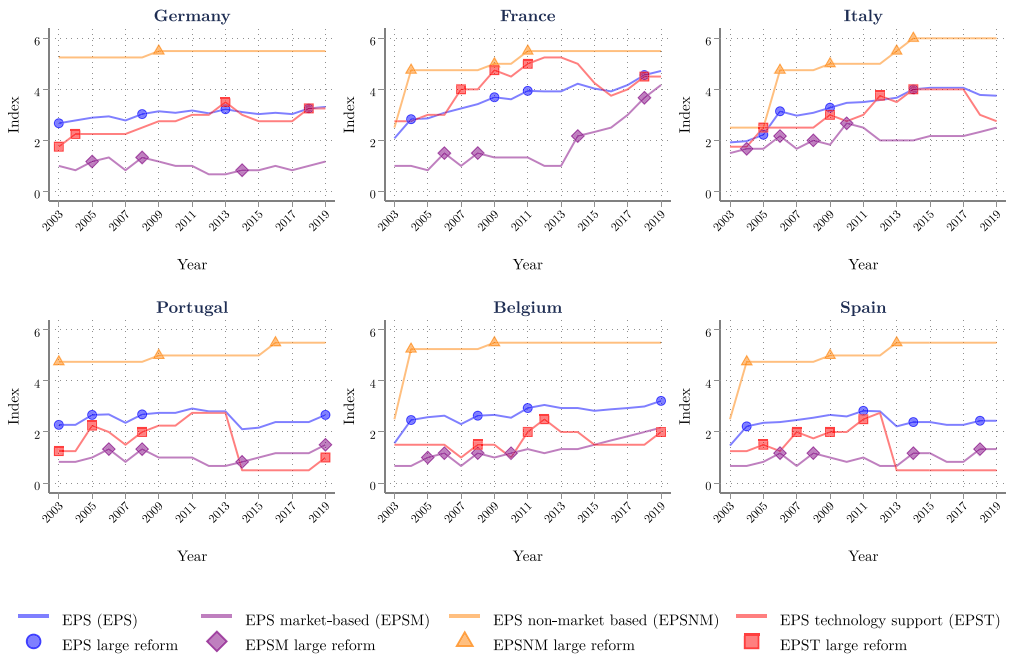

Fig. 2. Large EPS changes.
Note: Occurrences of large changes in the EPS sub-indicators in the six countries used in the firm-level analysis. A change is considered large if it is among the top 25% of the country's EPS change distribution.

remove potential asymmetric effects of policy softening. In particular, we consider the annual changes of the overall index and its sub-indicators and replace all negative changes with zero.[3]

In addition, we use a binary variable as our treatment which takes the value one if the increase in stringency is among the largest 25% of changes. Thereby, we remove all negative and small changes and focus on the salient regulatory changes where reactions by firms are expected to be more likely. The figure below (Fig. 2) shows these large reforms for each sub-indicator in each of the six countries in our data set, along with the year in which they occurred.[4] While regulatory reforms are usually not random, the changes in EPS indicators can be interpreted as independent (from past changes) and stationary shocks. We test for serial correlation of the positive and large changes in the EPS indicators using the bias-corrected (Born and Breitung, 2016) Q(p)-test and find no serial dependency for both specifications. The only exception is the serial correlation of positive changes in the technology support sub-indicator, which might stem from the serial correlation of GDP used as denominator to R&D subsidies. To further reduce the risk of endogeneity, we include lagged EPS changes in our regressions and exclude directly affected energy firms from our sample.

3.2. Firm-level data

Our firm-level analysis is based on firm data from six euro area countries – Belgium, France, Germany, Italy, Portugal and Spain - covering the period from 2003 to 2019. We are using the Bureau Van Dijk's ORBIS and the European Committee of Central Balance Sheet Data Offices' BACH (Bank for the Accounts of Companies Harmonized) data sets, which report balance sheet and profit-and-loss data for both listed and unlisted companies (BACH, 2015). A well-known concern regarding firm data is the lack of representativeness for some sectors and countries. Therefore, we use historical vintages of ORBIS and organise the data to improve the representativeness and reduce the sampling bias, as explained in Kalemli-Ozcan et al. (2015).[5] Additionally, for the five countries where iBACH data is available – France, Spain, Italy, Portugal and Belgium – the ORBIS and iBACH data sets are merged to improve coverage. Whenever a duplicate firm is observed, we keep the one from ORBIS. Data for Germany is retrieved only from ORBIS. We do not weight the data as the number of firms per country-sector-size-year cell is only available from 2009 onwards in the Structural

[3] Less than a fifth of the changes are negative (24% market, 0% non-market, 19% tech support) and these are spread across all countries. The results of Section 5 are robust when negative changes are not set to zero.

[4] At EU level (affecting all six countries), major market policy reforms are the introduction of the ETS (2005) and large changes in certificate prices (2006, 2018). An EU-wide non-market regulation to limit sulphur came into place in 2009. Country specific reforms are, among others, higher solar energy subsidies in Germany (2004), Portugal, Italy (2005) and France (2006–2012), NOx and PM emission limit reductions in Portugal (2003), Belgium, Spain and France (2004) or the introduction of CO_2 taxes in Spain and France (2014).

[5] Kalemli-Ozcan et al. (2015) argue that following their guidelines, there is no need to re-weigh the data to obtain nationally representative firm-level data sets.

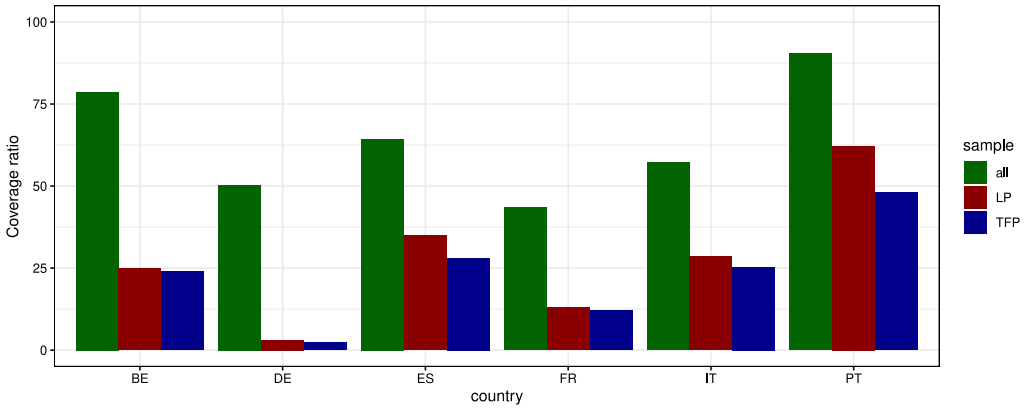

Fig. 3. Firm data coverage.
Note: Comparison of the number of firms in the data to the actual number of firms per country according to Eurostat. The bars "LP" (labour productivity) and "TFP" (total factor productivity) show firms with productivity information, while "all" captures the overall number of firms in the sample. Numerical values are shown in Table A.1.

Business Statistics from Eurostat and Bajgar et al. (2020) report that weighting does not solve potential representativeness issues. By restricting the sample to the best-covered European countries, imputing value added and focusing on firms above 10 employees the Orbis data are broadly representative (Bajgar et al., 2020). By including iBACH data, we improve the coverage of small firms.

We restrict our analysis to non-financial and non-governmental sectors, and remove firms in the mining, energy and real estate sectors (NACE Rev. 2 codes C to *N* except D, K and L). The included sectors are considered representative in the literature and it is possible to obtain reliable TFP values. Moreover, we excluded the energy sector to prevent that our results are driven by firms which are directly affected by the reforms captured in the EPS (in contrast to most other firms).[6] For the construction of our data set, we closely follow Kalemli-Ozcan et al. (2015) and Gopinath et al. (2017) to pursue a standard cleaning procedure. In particular, we keep only unconsolidated accounts and remove sole proprietorships,[7] and we remove negative value added and inconsistent balance sheet or income statement relations, including those with negative asset holdings.

Furthermore, we remove firm-year observations with less than one employee to be able to calculate TFP and LP, and we only keep firms with at least two consecutive years of reporting to be able to create growth rates. Finally, all balance sheet variables are trimmed at the 1st and 99th percentiles to limit the influence of outliers. Further, details on the data preparation can be found in the Appendix. The resulting data set covers three million firms (2.5 million with sufficient information to estimate Total Factor Productivity) with, on average, 10 years of information each. These are more than 22 million (18 million with TFP) firm-year observations from the covered six euro area countries. This represents about one-quarter of all firms with employees operating in the analysed sectors, but with important country differences: the coverage ratio ranges from 50% in Portugal, taking into account only firms for which we can estimate TFP, to less than 3% in Germany (Fig. 3).

Labour Productivity is computed as the real value added of the firm divided by the number of employees, where nominal value added is deflated with the 2-digit sector value added deflator. (Revenue) Total Factor Productivity takes into account the simultaneity bias arising from the fact that firms choose the optimal quantity of inputs given available information (to the firm, but not to the econometrician) on firm productivity and it is estimated following Ackerberg et al. (2015). See the Appendix for more details.

3.3. Emission estimation

One of the novel features of this paper is the estimation of CO_2 equivalent emissions for all firms in our sample using machine learning. Unfortunately, the availability of firms' emission data is very limited and biased towards large firms. However, financial and non-financial information can be used to infer CO_2 equivalent emissions. This effectively fills the data gap for firms which do not report emissions in order to support the analysis. Moreover, this synthetic data set can be used to monitor the exposures of the markets to polluting firms in an extensive number of other applications. The algorithm relies on emissions data provided by commercial providers for a sample of large listed firms, associated with the balance sheet information of those companies. The relevant quantities can then be predicted for non-reporting firms by applying the observed statistical relationships between the CO_2 equivalents emitted, the sector, the country and their financial information.

[6] Due to the small number of energy firms in the sample, the findings are, however, almost identical with and without energy firms.

[7] The inclusion of consolidated accounts would combine the financials of subsidiaries across different countries and industries and thus complicate comparisons across countries and sectors.

A	Re 1 2									
	0	1	2	3	4	5	6	7	8	9
0	168	88	30	31	18	8	5	7	6	3
1	56	117	54	22	21	13	10	3		2
2	29	68	87	44	28	42	17	7	11	
3	11	28	57	65	62	49	28	7	10	1
4	13	14	51	55	72	53	28	27	26	9
5	6	21	21	45	47	72	35	29	22	9
6	8	10	15	20	39	57	61	64	21	5
7	4	2	5	9	25	32	72	96	69	29
8	2	6	4	6	7	13	38	39	99	42
9	3	1	1	2	4	6	4	27	58	202

Fig. 4. Confusion matrix.
Note: Each row of the matrix represents the instances in an actual class while each column represents the instances in a predicted class. The matrix shows to which extent the algorithm "confuses" two classes. A large accumulation at the main diagonal show the fit of the estimation.

More specifically, we use Urgentem data on Scope 1 and Scope 2 CO_2 equivalent emissions[8] and link these with ORBIS/iBACH information to produce a sample of 35,000 firms. In order to simplify potential non-linearities in the relationship between emissions and financial information, we opted for creating ten emission quantity classes (bins) based on the distribution in our calibration sample, based on the initial distribution of available reported emissions. Once the model is fitted on the Orbis data set, the meaning of the bins should be treated as arbitrary categories. This allows us to rely on a wide classification set between high-polluters and low-polluters which gives us the possibility to test different definitions in the analysis. It also simplifies the calibration of the model. We use emissions rather than emissions intensity classes because most regulations and disclosure requirements relate to absolute levels of emissions and not to intensity. For the estimation of the data generation model we use a machine learning algorithm called Extreme Gradient Boosting (XGBoost), which is one of the most successful models used in machine learning in the past years, based on ensembled trees (Chen and Guestrin, 2016; Rokach, 2016). The model selects the regressors (via lasso algorithm) and finds the best non-linear patterns (tree) to estimate the dependent variable, then it averages them.

The confusion matrix (Fig. 4) shows how well XGBoost is able to estimate CO_2 equivalent emissions based on ORBIS/iBACH for firms never seen by the model before. The y-axis depicts the actual emission bins of our test sample, while x-axis shows the estimated bins. Since most observations in the confusion matrix are at or close to the main diagonal, it shows that actual and estimated emission bins are closely related and our algorithm works well. As we only need to split firms into high and low polluters for our empirical identification, the estimation accuracy is sufficient for our purpose. The bar plot in Fig. 5 depicts the mean of the absolute SHAP (SHapley Additive exPlanations) values for each regressor. The SHAP value captures how much a single regressor affected the prediction of an observation and this is summarized by taking the average over the sample (Lundberg and Lee, 2017). Employment, turnover,[9] tangible and intangible fixed assets as well as the (4-digit NACE) sector are the most important variables to determine emissions of a firm. It is difficult to assess ex-post the accuracy of classifications done on million of firms that do not report emissions but the relationships between sector, size and emissions follows our expectations (see Tables A.2 and A.3). As additional check, we estimated our model (in Section 4) only with large firms, comparable to the Urgentem firms, and the results are consistent with those of the overall sample.

4. Empirical strategy

Describing the causal impact of a regulatory reform over time poses various challenges, as described in the literature review. First, policy reforms create a dependency with business cycle dynamics and their impact could be heterogeneous reflecting macroeconomic or firm-specific conditions. Furthermore, policy and productivity changes could be jointly determined by many, sometimes unobserved, variables. Second and regarding environmental policies, Brunel and Levinson (2020) point to the problem of simultaneity, since not only environmental reforms affect macroeconomic outcomes but also macroeconomic conditions influence the implementation of reforms. Third, in a cross-country analysis we have to deal with different industrial compositions. Inferences can be incorrect, because even identical laws have different effects when the average manufacturer face different environmental costs across countries. Hence, in this section, we explain how we estimate the regulatory effects and how we deal with these concerns.

Our analysis builds on two layers. We start with the analysis of the impact of environmental policies on aggregate productivity growth (at the country-level) to capture the effects on the whole economy. In a second step, we explain our aggregate findings and deal with potential endogeneity concerns by looking at the effects on firm-specific productivity growth. Since we expect that effects of regulatory changes occur with some delay, we apply an econometric approach that allows to capture effects at different horizons. Local projections, introduced by Jordà (2005), calculate impulse responses (to policy reforms) directly by estimating regressions at

[8] The data capture direct emissions due to own production and indirect emissions due to purchased energy. CO_2 equivalents are used to compare and aggregate the emissions from various greenhouse gases: carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), hydrofluorocarbons (HFCs), perfluorocarbons (PCFs), sulphur hexafluoride (SF_6) and nitrogen trifluoride (NF_3).

[9] The level of emissions depends on the quantity of production, which we proxy with turnover.

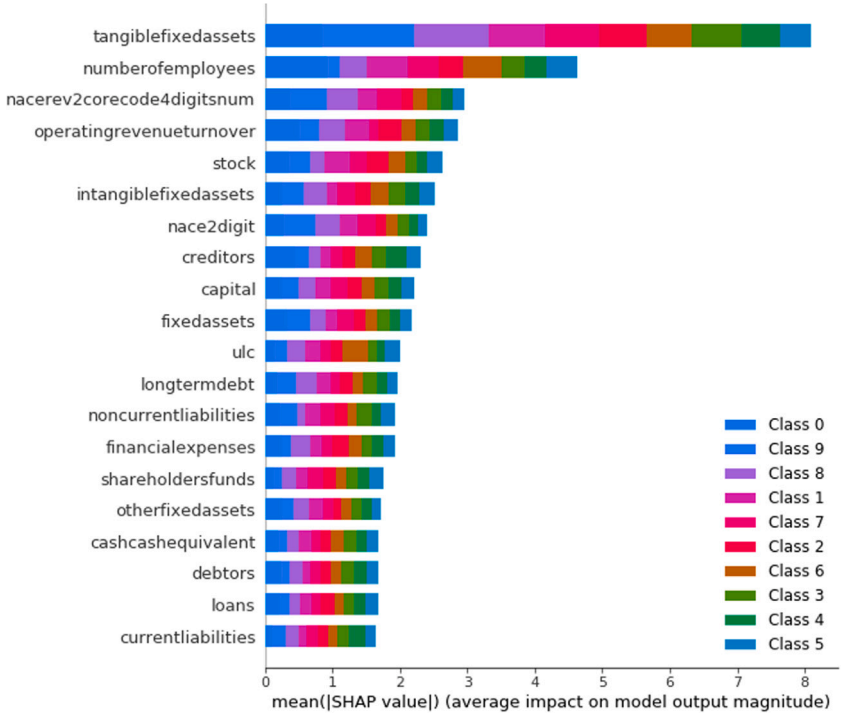

Fig. 5. Importance of balance sheet variables - SHAP values.
Note: Importance of variables in the machine learning algorithm, measured by the mean of the absolute SHAP value, starting with the most important and going to the least important.

each horizon of interest. This technique is based on sequential regressions of the dependent variable shifted forward in time onto its lagged values, while keeping the rest of the equation unchanged. Cette et al. (2020, p.7) describe the method as the "differences between two forecasts – the first corresponding to a situation with the shock and the second to the same situation without this shock". Compared to Vector Autoregressions, local projections are less prone to miss-specifications (Li et al., 2022) and more flexible regarding the analysis of non-linear and state-dependent impacts, while still able to deal with endogeneity issues (Bordon et al., 2018). More details on local projections can be found in the Appendix. We will apply this approach to capture the effects of environmental policy changes on productivity.

Our identification strategy is based on the idea that environmental policies affect firm productivity differently depending on the firms' exposure to environmental regulation. Due to different technologies, firms produce with different emission intensities which lead to heterogeneous adjustment needs in case of regulatory changes. Therefore, high-polluting firms are expected to face higher productivity losses in reacting to increased policy stringency compared to their low-polluting peers. We test this hypothesis to show the underlying mechanism that explains the intended direction and to exclude reversed causality (from productivity growth to regulatory changes).[10]

Thereby, we follow the approach popularized by Rajan and Zingales (1998) to describe the causal impact of country-level policies on industries or firms. They base their analysis on a pre-determined level of financial development (= policy), an indicator for financial dependency (= exposure), and industry and country fixed effects to test the underlying mechanism at play, and argue for the causal effect of financial development on economic growth. Beside many other applications, Albrizio et al. (2017) use the industry-level pollution intensity (= exposure) to show different effects of environmental reforms (= policy) depending on the regulatory exposure. In our empirical setting this approach implies that the environmental policy stringency indicator (= policy) is interacted with a pollution indicator specific to countries or firms (= exposure). We then test whether a high exposure has negative effects on productivity. While our emission indicator is better suited than those of former studies as it is firm-specific and varies

[10] Although endogeneity concerns are more elaborated at country-level, we also expect that countries are affected differently depending on their exposure and thus test the underlying mechanism.

over time, it is not continuous but limited to two (or ten) bins. However, different cut-offs among the emission bins and changing the exposure variable from two categories to ten bins do not change our conclusions.[11]

Beside the assumption that different technologies lead to variations in emissions and hence to different adjustment costs, the identification is based on the assumption that the exposure variable (emission bins) is exogenous to regulatory changes. Hence, it is crucial for this approach to use pre-determined (lagged) emission indicators that are unlikely to be affected by current changes in environmental regulation. In addition, we test whether an increase in regulatory stringency affects the probability of being a high-polluting firm and we find no significant impact. Table A.6 shows the effects of positive EPS changes on the emission indicator (logit regression) and the ten emission bins (linear regression). Firms being among high-polluters might reduce their pollution, but remain among high-polluters up to five year after the reform. In a similar way, most high polluting countries remain in this group over the whole period, although they introduced environmental policies. This supports the validity of our identification scheme.[12]

The dependent variable of our empirical model is the cumulative change in productivity, measured as total factor productivity or labour productivity, between year h after the reform and the pre-reform year, in percentage points (pp). The main regressor ($\Delta EPS_{i,t}^{j}$) is the positive change in the EPS index or the positive change in one of its sub-indicators j. We interact the policy index with a dummy (CO_2) indicating if the country or firm was among the top most polluting ones of its peer group in the year before the regulatory reform.[13] By using the change in the EPS index between $t-1$ and t, we limit the problem of simultaneity as it is unlikely that future TFP changes affect past regulatory changes.[14]

The local projection equation for our aggregate analysis is as follows:

$$ln(y_{i,t+h}) - ln(y_{i,t-1}) = \beta_1^h CO2_{i,t-1} + \beta_2^h \Delta EPS_{i,t}^{j} + \beta_3^h (\Delta EPS_{i,t}^{j} * CO2_{i,t-1}) + \gamma_1^h X_{i,t} + \alpha_{1,i}^h + \alpha_{2,t}^h + \varepsilon_{i,t+h} \qquad h = 0, 1, \ldots, 5 \tag{1}$$

where $y_{i,t+h}$ is the productivity variable (TFP or LP) of country i in year t; h represents years after the reform. $\varepsilon_{i,t+h}$ captures the idiosyncratic error. X captures country controls including the cyclical position of the country's economy, country TFP frontier growth, labour market regulation, startup costs, governmental R&D expenditure and the level of economic development, following Albrizio et al. (2014) if the variables are available. The last four controls are included as first lags (before the reform) so that they can be interpreted exogenously and not as an outcome of the environmental policy change. The cyclical position of the country is measured by the output gap and controls for business cycle dependencies of productivity growth and new regulations. The TFP frontier growth proxied by the US TFP growth captures common trends and technology spillovers among countries or firms. The employment protection legislation indicator and startup costs, which are provided by the OECD and the World Bank respectively, approximate supply side policies at the labour and product market. Governmental R&D expenditure controls for another potential omitted variable bias as it influences both firm productivity as well as decisions on green subsidies. A detailed description of the variables can be found in the Appendix (Tables A.4 and A.5). In addition, we use country fixed effects (α_i) to capture the different institutional settings across countries, as well as differences in corruption and climate that are potentially relevant to productivity. We include time dummies (α_t) to capture developments specific to a year but common to all countries like the ECB's monetary policy or the occurrence of the global financial crisis. Regarding the concern of industrial composition differences across countries, we add the industry share as control variable and see no changes in the baseline results.

At the firm-level, the dependent variable $y_{f,t+h}$ is the productivity variable (TFP or LP) of firm f in year t. While the policy indicator $\Delta EPS_{i,t}$ is specified at country level, we identify the past pollution $CO2_{f,t-1}$ on firm level through our machine learning algorithm. Beside the macro controls X that are also used in the aggregate specifications, we add lagged firm characteristics Z (before the reform) such as age, size, return-on-assets and distance to sector frontier as control variables. We replace the US TFP growth by the TFP growth of the top 5% most productive firms per sector and year to measure the TFP growth at the frontier.

$$\begin{aligned} ln(y_{f,t+h}) - ln(y_{f,t-1}) = \beta_1^h CO2_{f,t-1} + \beta_2^h \Delta EPS_{i,t}^{j} + \beta_3^h (\Delta EPS_{i,t}^{j} * CO2_{f,t-1}) + \gamma_1^h X_{i,t} + \gamma_2^h Z_{f,t} + \\ + \alpha_{1,t}^h + \alpha_{2,s}^h + \alpha_{3,f}^h + \varepsilon_{f,t+h} \qquad h = 0, 1, \ldots, 5 \end{aligned} \tag{2}$$

In the firm-level analysis, we replace country FE with more granular (4-digit) sector and firm FE (α_s, α_f) to control for unobserved heterogeneity across sectors and firms. The combination of sector, firm and time FE reduces a potential omitted variable bias drastically. In order to control for potential anticipation effects of new policies, we additionally include next year's EPS change in the firm-level specification, assuming perfect foresight. As this is a strong assumption and the inclusion reduces the magnitude of our estimated effects only slightly, we decided not to include it in the baseline specification. Overall, we use a broad set of macro and firm controls as well as firm, sector and year fixed effects to reduce the omitted variable bias to a minimum. Analysing granular firm data, using lagged controls, looking up to five year into the future and testing the direction of the regulatory effects according to Rajan and Zingales (1998), we can reliable reject the possibility of reversed causality.

[11] While our identification approach is different to the original (Bartik, 1991) instrument, it still follows the structure of a shift-share instrument, where the policy change represents the aggregate shift component and the exposure dummy represents the granular share component. Following the same logic, we try to make both components as exogenous as possible. The exogeneity of already one component is sufficient for the identification (Goldsmith-Pinkham et al., 2020).

[12] Moreover, we evaluate the correlation between emissions and TFP *growth* and find low and statistically insignificant coefficient values: 0.0008 (emission bins) and 0.0006 (emission dummy) - indicating a weak/no relationship between our exposure measure and the dependent variable.

[13] Interacting the policy index with all ten firm emission bins (assuming it to be a continuous variable) or with the initial (= first observed) emission dummy brings our analysis closer to the setup used by Albrizio et al. (2017) and leads to robust results as shown by Figs. A.7 and A.8.

[14] To test for this form of endogeneity, we regress changes in the EPS index on past country TFP growth. The lags of TFP growth do not predict the changes in the EPS index independent of the inclusion of control variables.

The estimation of the local projections is done by using Stata's reghdfe command on the pooled sample of 15 countries in the aggregate analysis or about 2.5 million firms from six euro area countries in the firm-level analysis. The standard errors are clustered at country or firm-level respectively to allow for fully flexible time series dependence in the errors within each block.[15] Note that the standard errors of our firm-level data may be underestimated because we cannot account for the potential estimation uncertainty of the emission bins and TFP in the regressions. However, given the broad categories used for the emission bins and the large number of firms included to estimate TFP, this should not be a significant problem. The set of robustness results, including different emission bin cut-offs (testing emission bin uncertainty) and the use of labour productivity (as an alternative measure of TFP), supports our case.

5. Results

In this section, we present the impact of changes in the environmental policy stringency on productivity growth. First, we describe the results at the aggregate level for a pooled sample of 15 countries to motivate our identification strategy and to get some intuition about regulatory effects in the euro area. Second, we use firm-level data to control for potential sources of endogeneity and to explore the heterogeneity in the regulation impact along several dimensions. By analysing the aggregate country data, we consider the full firm population, including entry and exit, and the relative importance of effects across firms weighted by their size. This analysis may, however, suffer from aggregation and endogeneity bias and is therefore mainly correlative. The firm-level analysis can address these problems, but has limitations in terms of population representativeness and tracking firm entry and exit dynamics.

5.1. Country-level analysis

Fig. 6 shows the impulse response functions (IRF) of the country-level analysis, using aggregate TFP growth as dependent variable. Results regarding the impact of environmental regulation on aggregate labour productivity are similar and shown in the Appendix (Fig. A.4). The orange lines show the mean responses to a one percentage point EPS increase over the five year horizon and the grey bands depict the respective confidence intervals around the point estimates. While the dark grey area represents the 68% confidence bands, the light grey area show the 90% bands. In the left column of Fig. 6 we show the productivity responses of the countries with the lowest greenhouse gas (GHG) emissions per capita (bottom 50% of the sample). We contrast them to the responses of the high polluting countries (top 50%) in the right column.[16] In line with our hypothesis that high polluters are more vulnerable to changes in the stringency of environmental regulation, we find that aggregate TFP of highly polluting countries is affected negatively and more strongly than that of less polluting countries.

Based on our aggregate results, we do not find support for the strong Porter hypothesis. While for the low-polluting countries the productivity effects of changes in the EPS index and its sub-indicators are close to zero or slightly positive, the regulatory impact on aggregate TFP growth of high polluters is significantly negative at the five year horizon (at least at the 68% confidence level). However, the negative effects on high polluting countries arise with different patterns for each sub-indicator. An increase in the stringency of the technology-support policies leads to a decline in productivity in the first two years which reverses afterwards to some extent. The effect is, however, significantly negative until the end of the horizon. In contrast, the effects of the market-based policy index and the non-market based policy index are muted in the short-run and show a declining trend only in the second half of the five year horizon, which drives the development of the overall EPS index. Hence, we do not find general support for the idea that a tighter environmental regulation leads to aggregate productivity increases that outweigh the costs of implementation, neither in the short-run, nor at the end of the projection horizon.

In order to evaluate the economic relevance of the regulatory impact, it is important to set the EPS shock size and the medium-term TFP development in perspective. The regulatory stringency shocks include small changes due to increases in CO_2 prices in the ETS or increases in green subsidies. At the same time, it also contains large changes caused by major reforms and price swings. While we plot the responses to a one percentage point increase in the EPS indicator, the mean change over all years and countries is around four pp for the overall indicator as well as its sub-indicators. A one standard deviation (SD) shock ranges between 13 pp for the overall indicator and 33 pp for the technology-support indicator. Hence, the effects of environmental policy reforms on TFP are larger in magnitude than depicted by the scale used in Fig. 6. A one SD shock to the overall EPS indicator would result in an expected TFP decline of 2.8 percentage points after five years in high polluting countries. This is a large effect given the mean annual aggregate TFP growth rate of 0.48% (2.4% over five years) in all countries or 0.28% (1.4% over five years) in high polluting countries.

As we cannot rule out endogeneity issues completely, we repeat the exercise with a different kind of regulatory shock that is less persistent and hardly predictable. Instead of a time-series of positive changes in the EPS index, we use a dummy that equals one if a salient change takes place that is among the largest 25% of changes in a country within our period of observation. Fig. A.5 in the Appendix shows the respective IRFs. While the effects are slightly more positive for the low-polluters and generally larger, the

[15] Given the national EPS indices and national institutions in charge of collecting the firm data, clustering SE at the country level would also be appropriate in the firm-level specification. Due to the low number of clusters (6) and the more relevant correlation among firm observations, we decided to use firm clusters. Country cluster SE estimates are robust and can be found in the Appendix , Fig. A.9.

[16] Belgium, Germany, Estonia, Finland, Ireland, Luxembourg and the Netherlands are in all years among the countries with above median pollution intensity, Greece in 8 years, Austria in 3 years. The distribution of average tons of GHG per capita is shown in Fig. A.3. We opt for using emissions per capita over emissions per GDP due to a lower risk of endogeneity, while it still controls for the size of the country.

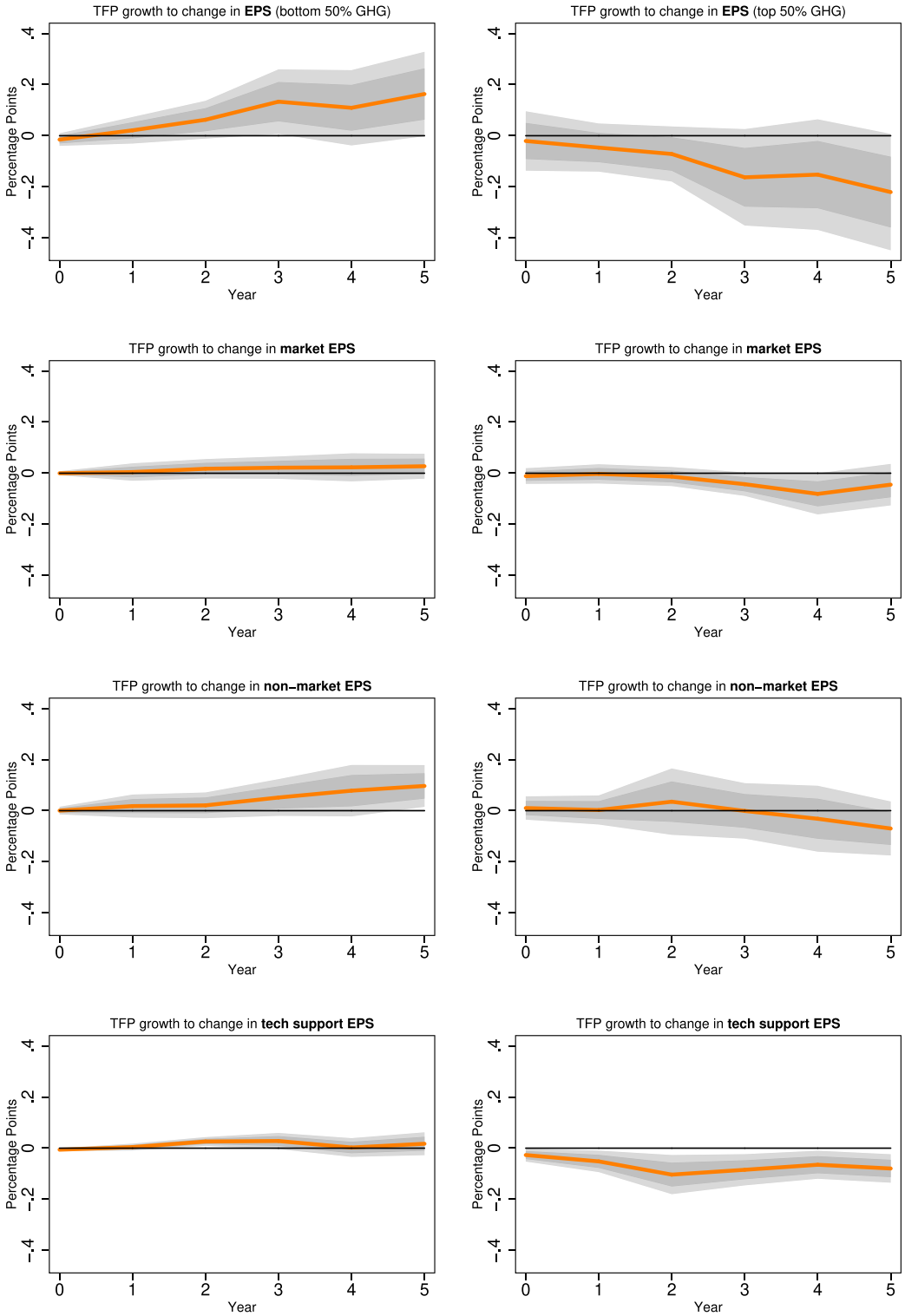

Fig. 6. Impulse response functions – country level.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive changes) over 5 years. Left column: bottom 50% polluting countries, right column: top 50% polluting countries. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

difference for the high-polluters is the same as before and the overall pattern is remarkably similar. Also using positive and negative EPS changes and looking up to ten years ahead (with respective larger uncertainty), does not alter our conclusions (see Fig. A.6). There is substantial heterogeneity among countries and their responses depend on their pollution intensity. Nonetheless, aggregate responses mask heterogeneous patterns across firms operating in a given country and sector as indicated by the large confidence bands. Moreover, there are still remaining concerns about potential endogeneity at the aggregate level. For these reasons, the next section replicates the analysis at the firm-level.

5.2. Firm-level analysis

Granular firm-level data help to reduce aggregate uncertainty, to limit endogeneity concerns and to explain heterogeneity in the regulatory impact on productivity. Therefore, we present the productivity responses of up to three million firms to positive changes in the EPS indicators similar to the country-level analysis, before we explore the heterogeneity across firms in the next section in more detail.

Fig. 7 shows the IRFs of firm productivity to a one percentage point increase in the EPS indicator and its sub-indicators. The blue lines represent the estimated coefficients β_2 and β_3 of Eq. (2) over a five year horizon and the grey areas represent the corresponding confidence intervals (68% and 90%). The significant differences between the left and right column confirm the direction of the relationship between regulatory reforms and TFP growth from regulation to productivity. As at the country-level, those firms that are highly polluting and hence more exposed are affected differently and to a larger degree than low-polluting firms. In particular, we compare firms that are assigned to the bottom four emission bins by our machine learning algorithm to those assigned to the top six bins by our machine learning algorithm (as described in Section 3.3). As there are less high polluting firms than low polluters, we decided for this unequal split, but robustness checks with different splits show similar results (see for example Fig. A.10 for the split top 4 vs bottom 6). Fig. A.7 leads to the same conclusions when we evaluate the impact on the firms in the bottom bin and the effect of moving to a higher bin, using a linear interaction term instead of a dummy variable in Eq. (2). The large number of low polluting firms is also a reason for the more precise estimation of the impulse responses in the left column. The confidence bands are so tight that they are not clearly visible in the plots. In this specification, we make use of the granularity of the firm-level data to control for sector and firm fixed effects so that we capture a broad set of reasons for different TFP growth rates and reduce the omitted variable bias to a minimum. Detailed regression results, including the coefficients of the control variables, can be found in Table A.7.

Qualitatively, the results are similar to the findings at country-level. The productivity of low-polluting firms is either not affected or slightly positively affected depending on the type of policy. In contrast, all policies affect highly polluting firms more negatively compared to the less polluting ones. The patterns of the various policies are, however, substantially different from each other. The productivity effects of technology-support policies are negative only temporarily before turning positive. Firms seem to need time to adapt to new conditions and make use of green R&D subsidies and feed-in-tariffs or auctions, in line with the findings of Aghion et al. (2020) on digitization. The lag between policy adoption and productivity stems from the processes inside the firm that have to evolve to unleash the productivity potential in the new regulatory framework. Both low-polluting as well as high-polluting firms benefit from subsidies in the long-run, while the latter go through a more painful transition.

In contrast, the non-market based measures have no immediate impact but become more costly over time. Non-market based instruments generally offer less flexibility to firms and we observe a persistent impact on exposed firms that takes a long time to compensate.[17] Finally, market-based policies have a significantly negative impact on firm productivity over the whole horizon, but at a less negative level than the non-market based policies. This is in accordance with the 'narrow' Porter hypothesis that states that market based measures are more productivity friendly than non-market based measures. However, in contrast to the 'strong' Porter hypothesis, the market based tools still have negative productivity effects on average. Since all three sub-indicators have an averse impact on firm productivity of high polluting firms (at least temporarily), it is no surprise that the overall indicator points in the same direction. While the negative short-run impact is driven by market based and technology support measures, the negative effects at the end of the horizon is dominated by non-market measures.

In accordance with our findings at the country-level, we have to reject the strong Porter hypothesis as soon as we consider pollution differences across firms. The average polluting firm exposed to environmental regulation faces a significant decline in productivity growth after changes in the stringency of environmental regulation. This is in contrast to Porter and Van der Linde (1995, p. 105) who claim that "[a]s the many examples [...] suggest, the opportunity to reduce cost by diminishing pollution should thus be the rule, not the exception". Our findings are underpinned by the very similar responses of labour productivity to EPS shocks as shown in the Appendix (Fig. A.11). Also controlling for a potential attrition bias, by analysing only firms with six observations or more, does not change our results and confirms their robustness.[18]

Again, we have to put the size of the regulatory impact in Fig. 7 in perspective. While we plot the responses to a one percentage point increase in the EPS indicator, the mean change is around four pp for the indicators. As mentioned before, a one standard deviation shock ranges between 13 pp for the overall indicator and 33 pp for the technology-support indicator, so that

[17] The peak in the fourth year of the IRF to EPS and non-market EPS changes is driven by large non-market policy reforms in Italy in 2006 (legislative decree 152/2006 with the aim to reduce SO_X, NO_X and PM) that doubled the non-market stringency level. Turning this effect off leads to similar but more muted results without a peak.

[18] Unfortunately, entry/exit dynamics cannot be captured completely, as the incorporation date information in ORBIS (iBACH) suffers measurement issues. In addition, the firm may remain active even if it drops out of the sample.

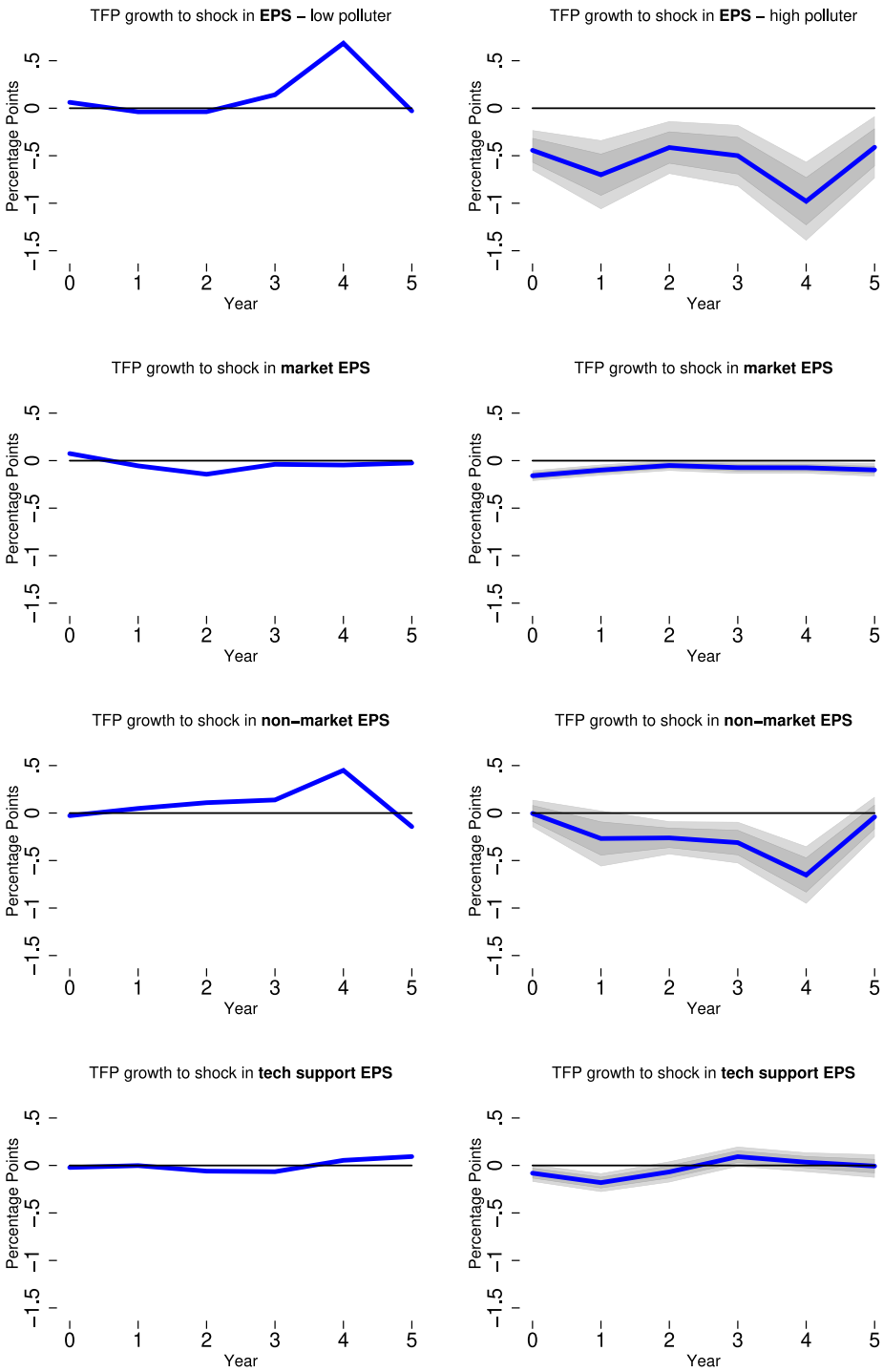

Fig. 7. Impulse response functions – firm-level.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive changes) over 5 years. Left column: bottom 4 bins of polluting firms, right column: top 6 bins of polluting firms. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

a multiplication of the shock size by ten (or more) is necessary to capture the impact of large environmental reforms. A one SD EPS shock reduces firm TFP growth of high polluters by four pp after five years on average. Given the median annual TFP growth of 2.6% among high polluting firms, a one SD EPS shock reduces the TFP growth by one-third over five years. In contrast to the overall and the non-market policy indicator, the sub-indicator for market based policies shows smaller negative effects on polluting firms. After five years the TFP growth declines by 2.6 pp to a one SD market EPS shock, which can be compensated by an average year of TFP growth. The median TFP growth of low polluting firms is significantly smaller (1.2%), but it is not negatively affected by more stringent environmental regulations. These findings are robust to the application of positive *and* negative changes in the EPS indices as shown by Fig. A.13. Using our dummy that equals one if a salient change takes place as our regulation indicator, we also find robust results, but with different patterns for some sub-indicators (see Fig. A.12). Similarly, TFP growth declines by four pp to large overall EPS changes after four years, and the negative impact of the technology support measures is only temporary as with the baseline reform indicator. However, the impact of the non-market policies becomes more temporary and large increases in market based policies seem to harm firm TFP growth relatively more.

While a detailed analysis of all potential spillover effects across firms and countries is out of the scope of this paper, we check the influence of European policies and general stringency changes across Europe. For this exercise, we use the country-specific EPS change and its interaction with the emission indicator in our local projections and add the average EPS change of the other five countries and the respective emission interaction. So, we can evaluate the impact of international regulations on high-polluting firms in addition to the national reforms. We find that country-specific reforms as well as reforms from abroad have a negative impact on polluting firms compared to low-polluting ones, however, with differences across policies. The negative effect of market policies stem exclusively from international reforms. This is not surprising since most of the observed variation in the EPS comes from the introduction and price changes of the ETS and many countries do not have own CO_2 taxes (or at a low level). In contrast, the negative non-market effects are mainly related to national reforms, which are less coordinated internationally.

5.3. Heterogeneity in firms' responses

Finally, we want to explore potential heterogeneities among firms, and in particular whether all high polluting firms are affected by more stringent environmental regulation in the same way. Therefore, we modify Eq. (2) to include a triple interaction effect to capture non-linearities in the regulatory shock on high polluting firms' TFP growth:

$$\begin{aligned} ln(y_{f,t+h}) - ln(y_{f,t-1}) = \beta_1^h CO2_{f,t-1} + \beta_2^h \Delta EPS_{i,t}^j + \beta_3^h (\Delta EPS_{i,t}^j * CO2_{f,t-1}) + \beta_4^h (\Delta EPS_{i,t}^j * CO2_{f,t-1} * T_{f,t-1}) + \\ + \gamma_1^h X_{i,t} + \gamma_2^h Z_{f,t} + \alpha_{1,t}^h + \alpha_{2,s}^h + \alpha_{3,f}^h + \varepsilon_{f,t+h} \end{aligned} \tag{3}$$

where $T_{f,t-1}$ is a standardized lagged variable which we use to differentiate among high polluting firms. We standardize the variables to ease the interpretation of variables with artificial measurement units (e.g. TFP) and to make the coefficients comparable to each other. As for the emission bins (CO_2), we use the lagged value to keep it unaffected by the current EPS shock. The local projections also include the other two interaction effects $EPS * T$ and $CO2 * T$, but they are not shown to increase readability.

Capabilities to deal with a changing market environment are crucial to deal with climate change and new environmental legislation. These capabilities are often associated with more experience and resources. Therefore, we first distinguish the firms by size (number of employees), age (years since incorporation) and the current level of productivity (TFP). To reduce the dimensionality of our results, we focus on (overall) EPS shocks and their impact after three years. The results of our heterogeneity exercise are depicted in Table 1. First, it shows that the pattern found in our baseline specification is robust to adding a triple interaction effect. The effect of an EPS shock on TFP growth remains positive and small for non polluting firms and significantly negative and large for high polluters. Second, considering the three firm characteristics size, age and productivity, we find that the number of employees and the TFP level are statistically relevant in explaining differences in the EPS's impact on productivity growth of polluting firms. Larger and more productive firms can handle regulatory changes more efficiently than the average high polluting firms. While a one SD larger firm (3300 employees more) reduces the negative impact of policy tightening by only 0.05 pp, a one SD more productive firm reduces the negative impact by 1.08 pp. Therefore, starting from a higher productivity level seems crucial to deal with regulatory changes successfully.

We next explore why large and more productive firms are better off after a change in the environmental regulation relative to peers. In particular, we want to know to which extent the innovativeness of firms and their access to financial resources play a role in mitigating the costs of regulatory shocks. As proxy for innovativeness we use the amount of patent families created in the past[19] and financial resources are captured by the equity ratio. According to our estimation results (Table 1), a larger equity ratio reduces the negative impact of an EPS shock on TFP growth. A one SD higher equity ratio (+31%) ameliorates the negative impact by a fifth. Similarly, having innovated in the past helps a firm to withstand negative effects of more stringent environmental policies. A one SD higher holding of patents (+37 patent families) reduces the negative impact experienced by high polluting firms by a tenth. Therefore, firms' innovativeness and access to finance are important mitigating factors of the short to medium-term negative impacts of stringent environmental policies.

This is supported by Fig. 8 where we distinguish between the impact of the EPS sub-indicators after three years (see Fig. A.14 for other horizons). Their effects on TFP growth of high polluting firms are vastly different when we compare firms at the 10th percentile of the size, equity or patents distribution (dark blue dot) to their counterparts at the 90th percentile (light blue bar).

[19] A patent family is a collection of patent applications covering the same or similar technical content. We use Orbis IP as our patent data source.

Table 1
Heterogeneity analysis – high polluting firms.

	Dependent variable: 3 year ahead firm TFP growth					
	(1)	(2)	(3)	(4)	(5)	(6)
$CO2_{t-1}$	3.152	2.590	3.534	2.031	2.318	4.905**
	(2.032)	(2.186)	(2.322)	(2.328)	(2.161)	(2.088)
EPS shock	0.141***	0.145***	0.136***	0.0560***	0.141***	0.141***
	(0.0057)	(0.006)	(0.0057)	(0.0104)	(0.0057)	(0.0057)
EPS shock * $CO2_{t-1}$	−0.501***	−0.593**	−0.401*	−0.391*	−0.550**	−0.495**
	(0.195)	(0.225)	(0.221)	(0.234)	(0.215)	(0.197)
EPS shock * $CO2_{t-1}$ * Employees		0.0524***				
		(0.01)				
EPS shock * $CO2_{t-1}$ * Age			−0.0076			
			(0.077)			
EPS shock * $CO2_{t-1}$ * TFP level				1.094**		
				(0.537)		
EPS shock * $CO2_{t-1}$ * Equity ratio					0.096**	
					(0.0177)	
EPS shock * $CO2_{t-1}$ * Cumulative Patents						0.0412***
						(0.0099)
Aggregate controls (Output gap, R&D, EPL, startup costs, TFP frontier)	Yes	Yes	Yes	Yes	Yes	Yes
Firm characteristics (age, size, ROA, TFP gap)	Yes	Yes	Yes	Yes	Yes	Yes
Firm and sector F.E.	Yes	Yes	Yes	Yes	Yes	Yes
Time F.E.	Yes	Yes	Yes	Yes	Yes	Yes
R^2 adjusted	0.57	0.57	0.57	0.42	0.57	0.57
N	6,081,287					

Note: Age, size and ROA are standardized in the local projections, TFP gap is not included in TFP level interaction regression. CO_2 represents top 4 bins of polluting firms. Parentheses contain firm-clustered standard errors. * < 0.1, ** < 0.05, *** < 0.01

While small polluting firms are negatively affected by more stringent non-market policies, large ones experience a TFP growth similar to low polluting firms. Stark differences in the effects of non-market policies can also be seen in case of different equity ratios and different patenting experience, however, at a lower level. The differences across firms are less relevant for market based policies. They have a negative but limited impact on the TFP growth of polluting firms with little differences across size, equity and patenting. It is important to see that technology support measures have already positive effects after three years, even for high polluting firms. The differences in innovativeness and financial resources matter only little for this policy type.

Eventually, we also want to point to two other dimensions of heterogeneity: differences in the regulation stringency level and sector differences. So far, the analysis only captures the change of the EPS indicators and not the EPS levels. While we (and the literature) argue that regulatory changes are more important than their levels in explaining productivity growth, also the policy stringency level can have economic effects. Including the lagged EPS indicator (in levels) as additional control variable in Eq. (2) does not alter our results presented in Fig. 7. However, splitting the sample into high and low EPS level countries (FRA/GER/ITA vs BEL/POR/ESP) shows that the negative effects of more stringent policies on polluting firms are not statistically significant at the end of the horizon for the less regulated countries. It seems that less strict regulation allows firms to adapt quicker to new regulations, potentially by making use of the experience and technology of more regulated firms. Distinguishing between the industry and service sector leads to similar results regarding their vulnerability to more stringent environmental regulations. High polluting firms in both sectors face productivity growth losses due to changes in the EPS indicators. Interestingly, there is a difference regarding the market based indicator. While higher carbon prices have significantly negative effects on polluting firms in the industry sector, the service sector faces only small negative effects, neither statistically nor economically relevant.

6. Conclusion

Given the acceleration of climate change, governments all over Europe increased their efforts in limiting the emission of green house gas emissions by implementing a broad range of policies. While stricter regulation is usually associated with a decline in competitiveness, there is the hope that well designed environmental regulation boosts innovation and thereby counteracts negative economic effects. This idea became known as the Porter hypothesis.

In this paper, we evaluate the strong Porter hypothesis by analysing the impact of changes in environmental policy stringency on TFP growth of countries and firms. We combine macro indicators with firm-level data and estimate firm-specific emissions to identify more and less exposed firms to regulation. By estimating panel local projections on firm and country level, including a broad set of covariates and fixed effects, we confirm that high polluters are affected significantly different than low polluters and thus the direction of the effect from regulation to productivity. In general, the productivity growth of countries with high pollution intensity

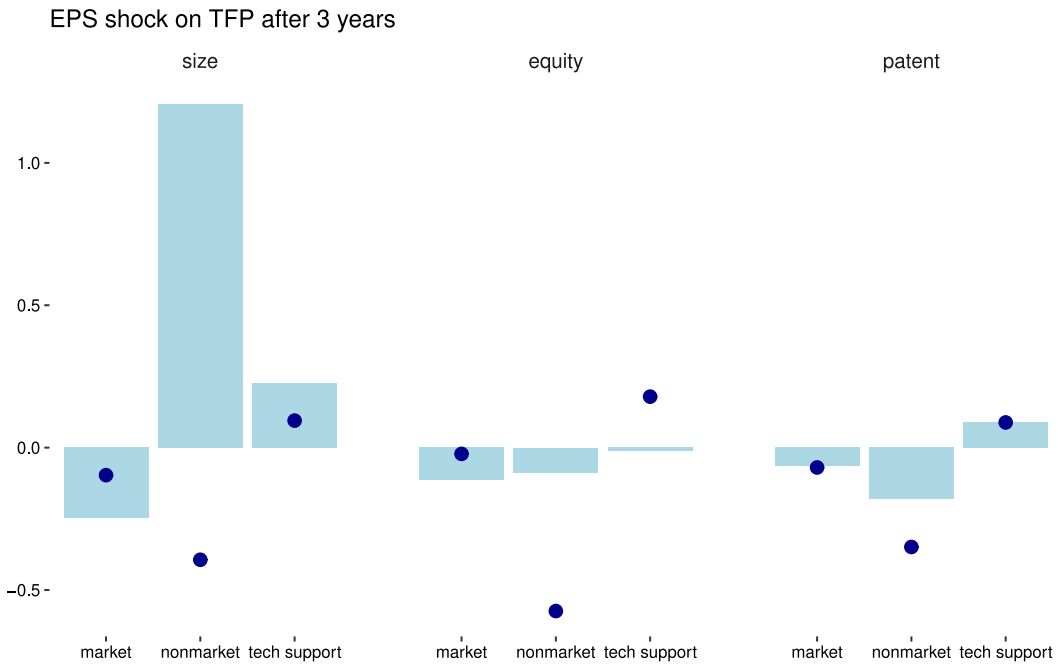

Fig. 8. Impact heterogeneity across firms.
Note: Cumulative TFP growth response to an EPS shock (positive changes) after three years. Dark blue dots represent effects on a firm at the 10th percentile of the size (number of employees), equity (equity ratio) or patent (cumulative patent families) distribution. Light blue bars represent the effects on firms at the 90th percentile of the respective distribution.

and firms with large CO_2 equivalent emissions are affected negatively by a change in environmental policies. Hence, we have to reject the strong Porter hypothesis that environmental regulation is productivity enhancing (in most cases). It is, however, important to distinguish between different policy options. Green subsidies have negative effects on TFP growth only during a transition period, before becoming beneficial. Market based measures (taxes) reduce TFP growth of high polluting firms, but less than non-market based regulation (emission limits). In addition, it matters whether firms have access to finance or are innovative: a higher equity ratio and more expertise with patenting reduce the negative impact of environmental regulation.

Potentially, market and non-market regulations could boost productivity of polluting firms in the long run in a way comparable to the positive effects of subsidies after three years. This is hinted at in a related paper which finds that more stringent environmental regulation increases significantly the number of green patents over a five year horizon after the regulatory change. If we believe that new innovations need time to be adapted to different production processes and, therefore, render productivity gains, the strong version of the Porter hypothesis could be confirmed over a longer time horizon. Given our limited period of data, we are, however, only able to look five years into the future. Using a longer time series, including firms from more countries and having better firm-specific emission data could further improve the analysis, but is left for future research. Up to now, we have to dash the hope that environmental regulation has no economic costs, but we also show that the costs on TFP growth are limited and by choosing the right policies they can be minimized. We also show that better access to capital and innovativeness can mitigate the short to medium-term costs of adjusting to a greener way of producing.

CRediT authorship contribution statement

Nicola Benatti: Writing – review & editing, Visualization, Resources, Methodology, Data curation, Conceptualization. **Martin Groiss:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Petra Kelly:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Paloma Lopez-Garcia:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

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We thank Giovanni Primativo for outstanding research assistance helping us to prepare the firm-level data.

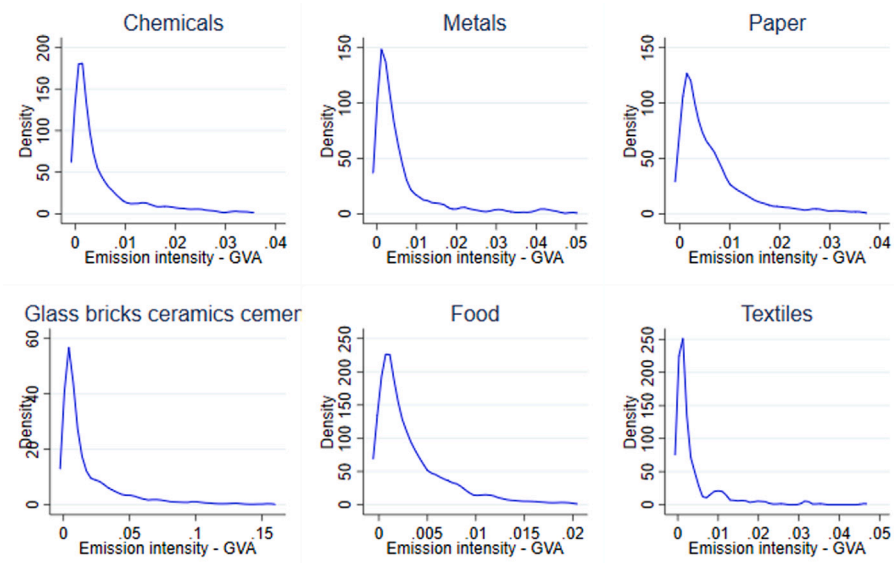

Fig. A.1. Within sector emission heterogeneity.
Note: Emission intensity distributions of selected sectors based on emission data from the European Trading Scheme (ETS) and output data from ORBIS. All years and countries are combined. The use of emission intensity instead of levels controls for the different size of the firms. GVA – gross value added.

Appendix

Firm-data preparation

Data preparation ORBIS + IBACH (following Kalemli-Ozcan et al., 2015):

1. Access current and historic Orbis vintages to increase coverage
2. Delete observations without BvD ID or account number
3. Link firms with correspondence table in case the BvD ID changes
4. Merge iBACH data and drop duplicates from iBACH
5. Drop duplicates in terms of ID and year
6. Drop observations with no financial information
7. Drop observations where country code and country letters in BvD ID do not match
8. Use unconsolidated accounts of the firms and remove sole proprietorships
9. Adjust all financial variables to be the same units and remove firms where it is not clear in which units numbers are reported.
10. Drop observations with missing: total assets & operating revenue & sales & employment
11. Drop observation if employment is larger than 2 million or turnover is above 500 billion € in any year
12. Remove the value of employment, sales, any costs, any asset or any loan if it is negative
13. Remove the value if EBITDA > 100 billion € or any asset > 4 trillion € or cost of employees = 0
14. Cleaning jumps for: operating revenue, tangible fixed assets, fixed assets, number of employees, costs of employees, added value, depreciation amortization
    (a) Remove the value if one observation is 50 times larger or smaller than its previous or next value (or replace all values with NA if a firm has no more than four observations of this variable)
    (b) Replace all values of a firm's variable with NA if there is a change by a factor of more than 500
15. Remove all values of cost of employees (from a firm) if costs per employee has changed more than ten times and number of employees has not fallen or increased by more than ten times within any year

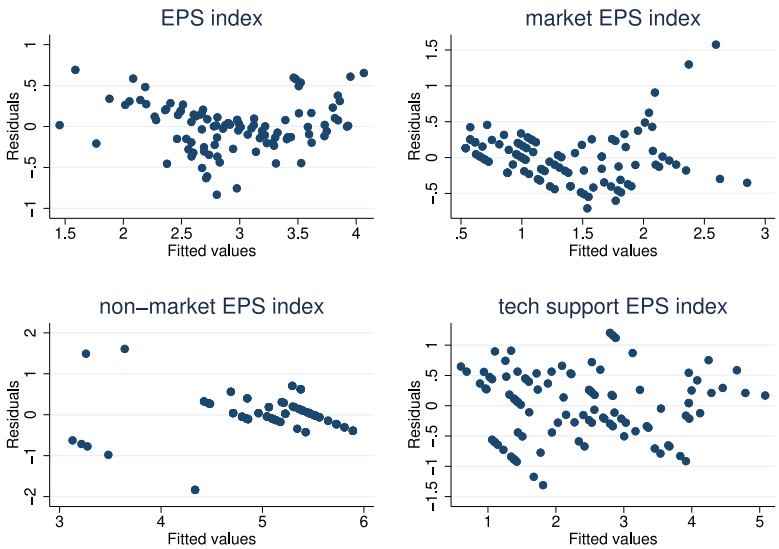

Fig. A.2. Residual plot – EPS indices.
Note: This plot shows the variability of the EPS indices after controlling for country and year fixed effects.

16. Remove all values of number of employees (from a firm) if turnover or cost of employees increased by more than ten times and number of employees fall by more than ten times in the same year
17. Nominal variables are deflated by country and sector specific value added deflator or capital deflator, respectively

Additional own adjustment for TFP/LP calculation:

1. Drop firms not in the sectors C to N and drop firms in the sector D, K or L.
2. Drop observations that do no report employment (employment < 1)
3. Drop firms with no consecutive observations (needed for growth rate)
4. Balance sheet variables are trimmed at the 1st and 99th percentile in each sector

TFP estimation

A number of different approaches exist for estimating Total Factor Productivity with Ackerberg et al. (2015), Wooldridge (2009), Levinsohn and Petrin (2003) and Olley and Pakes (1992) among the most popular ones. In this paper, Total Factor Productivity at the firm-level is estimated following the approach of Ackerberg et al. (2015) building on Levinsohn and Petrin (2003) to correct for the so-called "simultaneity bias". This bias occurs when the firm's productivity is unknown for the econometrician but known to the firm before choosing optimally production inputs. The output measure used in the analysis is firm-level value added deflated with 2-digit industry deflators from the STAN database of the OECD. Intermediate inputs are used in the inverted policy function estimated in the first stage. The coefficients of the production function are estimated in the second stage by GMM regressions at the 2-digit industry, pooling all countries together. We also follow the recommendations by Gal (2013) and address issues with the Orbis data such as the under-representation of small firms and missing data for key variables in certain countries that require imputation.

Although our policy measure (EPS changes) is measured at the country level and has no serial correlation, there might be remaining endogeneity between firms' TFP and changes in environmental regulation. Therefore, we follow the approach by Loecker (2013), who considers the role of exporting in the TFP estimation, and include our policy variable as endogenous variable in the TFP estimation procedure (Rovigatti and Mollisi, 2018). Given the nearly identical TFP values with and without considering EPS changes (correlation of 0.99999), we only report the results without including the EPS changes to keep the TFP measure directly comparable to former research. The similarity of the outcomes are no surprise, as Gal (2013) notes that correlations for growth rates between different TFP measures are high. This is also confirmed by Silveira et al. (2021), which finds similar results with different TFP estimates. Details on remaining biases and uncertainty in the TFP estimation approach are explained in Van Beveren (2012).

Table A.1
Firm data coverage — iBACH/ORBIS (2003–2019).

Country	Population	All obs	Ratio all	TFP obs	Ratio TFP	LP obs	Ratio LP
BE	4,968,914	3,907,666	78.64	1,189,084	23.93	1,238,359	24.92
DE	11,670,030	5,875,611	50.35	272,223	2.33	357,360	3.06
ES	16,895,364	10,866,459	64.32	4,745,566	28.09	5,904,770	34.95
FR	21,839,701	9,478,328	43.40	2,643,235	12.10	2,823,961	12.93
IT	17,877,928	10,205,926	57.09	4,532,361	25.35	5,115,723	28.61
PT	5,087,249	4,596,188	90.35	2,448,580	48.13	3,165,136	62.22

Note: Population figures by Eurostat (structural business statistics), observations from merged data set — ORBIS and iBACH.

Table A.2
Emission bin distribution — Firm size.

CO_2 bin	Range	Firm size (employees)					
		1–9	10–19	20–49	50–249	GE250	Total
0	(0, 8,535]	15,745,469	3,237,063	2,210,002	1,140,006	197,330	22,529,870
1	(8,535, 28,875]	19,016	1372	1780	5430	22,399	49,997
2	(28,875, 61,367]	2222	577	489	1433	1760	6481
3	(61,367, 120,071]	364	17	35	90	3129	3635
4	(120,071, 223,372]	2561	504	377	239	1022	4703
5	(223,372, 421,441]	37	1	1	16	599	654
6	(421,441, 832,987]	0	0	1	2	1107	1110
7	(832,987, 1,956,327]	0	0	0	1	244	245
8	(1,956,327, 6,933,000]	0	0	0	11	344	355
9	(6,933,000,]	476	87	66	17	305	951
Total		15,770,145	3,239,621	2,212,751	1,147,245	228,239	22,598,001

Note: The rows show the number of firms in each emission bin for each firm size class. GE = greater or equal. The range is denoted in tonnes of CO_2 emission equivalents per year.

Table A.3
Emission bin distribution — Sectors.

CO_2	NACE 2 sectors								
bin	C	F	G	H	I	J	M	N	Total
0	4,255,666	3,851,022	6,892,538	1,214,609	1,917,112	923,011	2,358,560	1,117,352	22,529,870
1	18,498	6254	11,180	4365	2815	1724	3467	1694	49,997
2	2108	867	1392	389	376	463	512	374	6481
3	1955	99	651	212	26	176	403	113	3635
4	907	544	1137	499	176	199	863	378	4703
5	360	23	45	115	1	78	27	5	654
6	880	12	116	64	5	29	3	1	1110
7	130	2	4	60	0	49	0	0	245
8	224	11	10	61	0	23	4	22	355
9	134	215	343	230	1	6	7	15	951
Total	4,280,862	3,859,049	6,907,416	1,220,604	1,920,512	925,758	2,363,846	1,119,954	22,598,001

Note: The rows show the number of firms in each emission bin for each NACE sector.

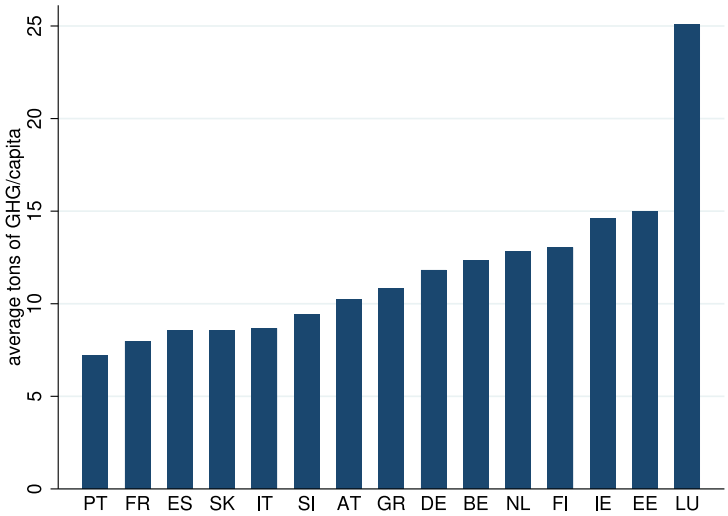

Fig. A.3. Greenhouse gas emission distribution.
Note: This graph shows the average tons of greenhouse gas emissions per capita over the period 2003–2019 for the 15 EU countries included in our aggregate analysis. It shows the similarity among the bottom 6 (PT, FR, ES, IT, SK, SI) and top 7 (LU, EE, IE, FI, NL, BE, DE) countries, with Austria and Greece being in the middle.

Local projections

The "local projection" method is a statistical and econometric technique used to analyse the relationship between economic variables over time. It was developed by Jordà (2005), Jordà et al. (2015) and is particularly useful for understanding how a particular variable (e.g., GDP, inflation, or productivity) responds to economic shocks or changes in other variables over time. They can help you uncover time-varying patterns and relationships that traditional methods may overlook. This flexibility is especially important when studying economic phenomena that are subject to change and variation over time, as it provides a more accurate and detailed analysis of the data.

By using local projections it is possible to directly calculate impulse responses by estimating regressions sequentially at each horizon of interest rather than extrapolating into increasingly distant horizons as done by vector autoregressions (VAR). Compared to VARs, local projections also have other advantages: first, local projections are easier to estimate, because they only consist of one linear regression; second, local projections also less prone to miss-specifications (Li et al., 2022); and third, they are more flexible regarding the analysis of non-linear and state-dependent impacts since interaction effects can be easily added. Yet, Plagborg-Møller and Wolf (2021) show that local projections and VARs estimate the same impulse responses when the lag structures are unrestricted.

Once an shock has been identified, impulse responses can be directly estimated using OLS regressions (Adämmer, 2019):

$$y_{t+h} = \alpha^h + \beta_h * shock_t + \gamma X_t + u^h_{t+h} \qquad h = 0, 1, \ldots, H-1, \tag{4}$$

where α^h is the intercept, X_t is a vector of control variables, and $shock_t$ is the identified shock variable. The coefficient β_h describes the expected response of y at time $t+h$ to the shock variable at time t. The impulse responses are the sequence of all estimated β_h. Robust standard errors are usually estimated using the approach by Newey and West (1987). Following the macroeconometrics literature on VARs, which often faces a small number of observations and difficulties in constructing confidence intervals due to serial correlation, different confidence bands are plotted. To see the likelihood shape and not just the 95%-threshold, also the 68% (1 SD) confidence level is suggested by Sims and Zha (1999).

As local projections disentangle the economic effects of policies from background shifts in the economic environment, local projections have been extensively deployed in particular in the context of evaluating fiscal policy impact (Auerbach and Gorodnichenko, 2012; Jordà and Taylor, 2016; Ramey and Zubairy, 2018; Born et al., 2019) and more recently for also in measuring the impact of product and labour market reforms (de Haan and Wiese, 2022; Bordon et al., 2018). To our knowledge our contribution is the first to apply it in an environmental policy context, with the exception of Eugster (2021) and Bettarelli et al. (2023) who apply it in country-level empirical studies, and who focus on renewable energy patents as the outcome.

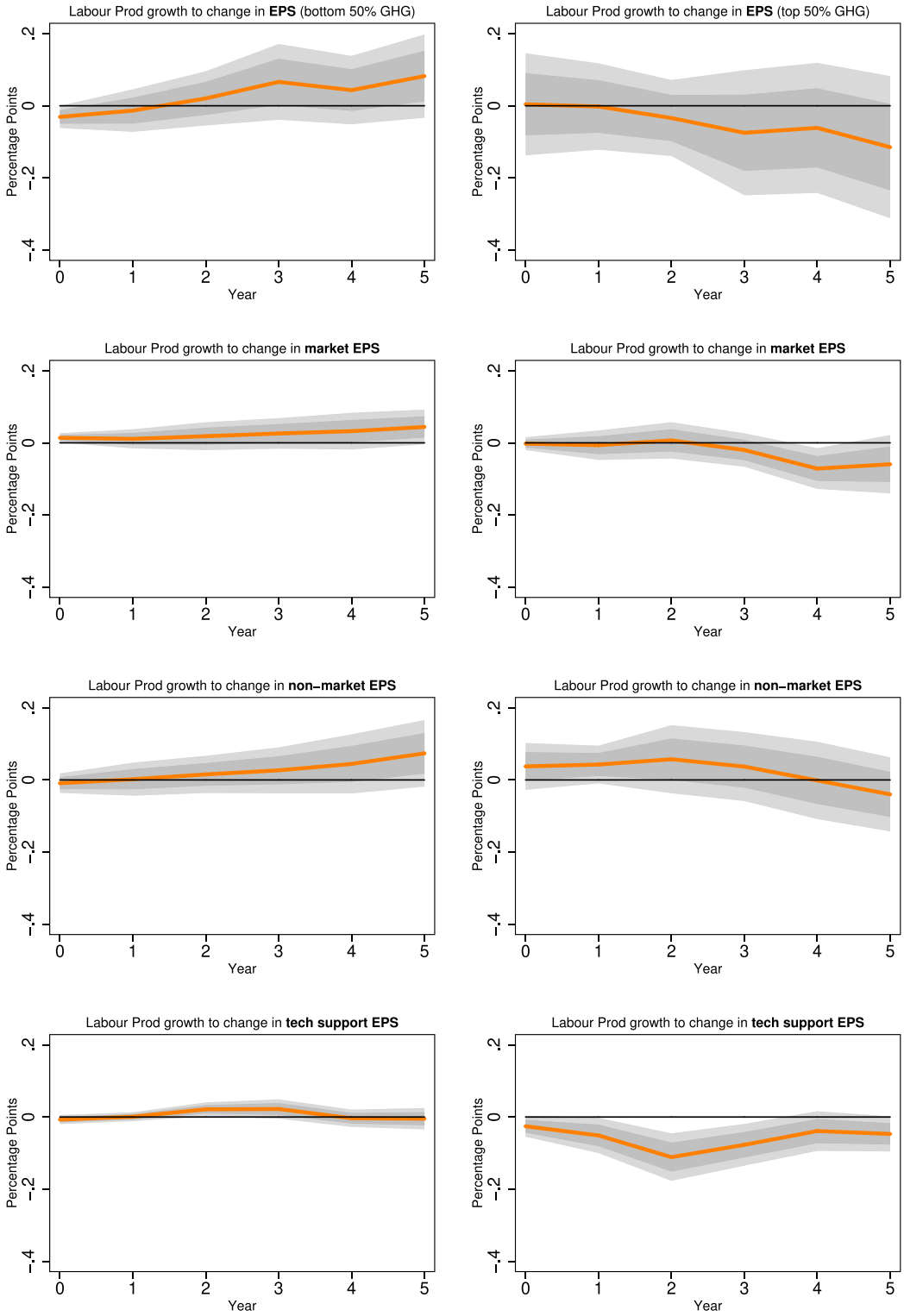

Fig. A.4. Impulse response functions – country-level – labour productivity.
Note: Cumulative impulse responses of labour productivity growth to 1 pp EPS shocks (positive changes) over 5 years. Left column: bottom 50% polluting countries, right column: top 50% polluting countries. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

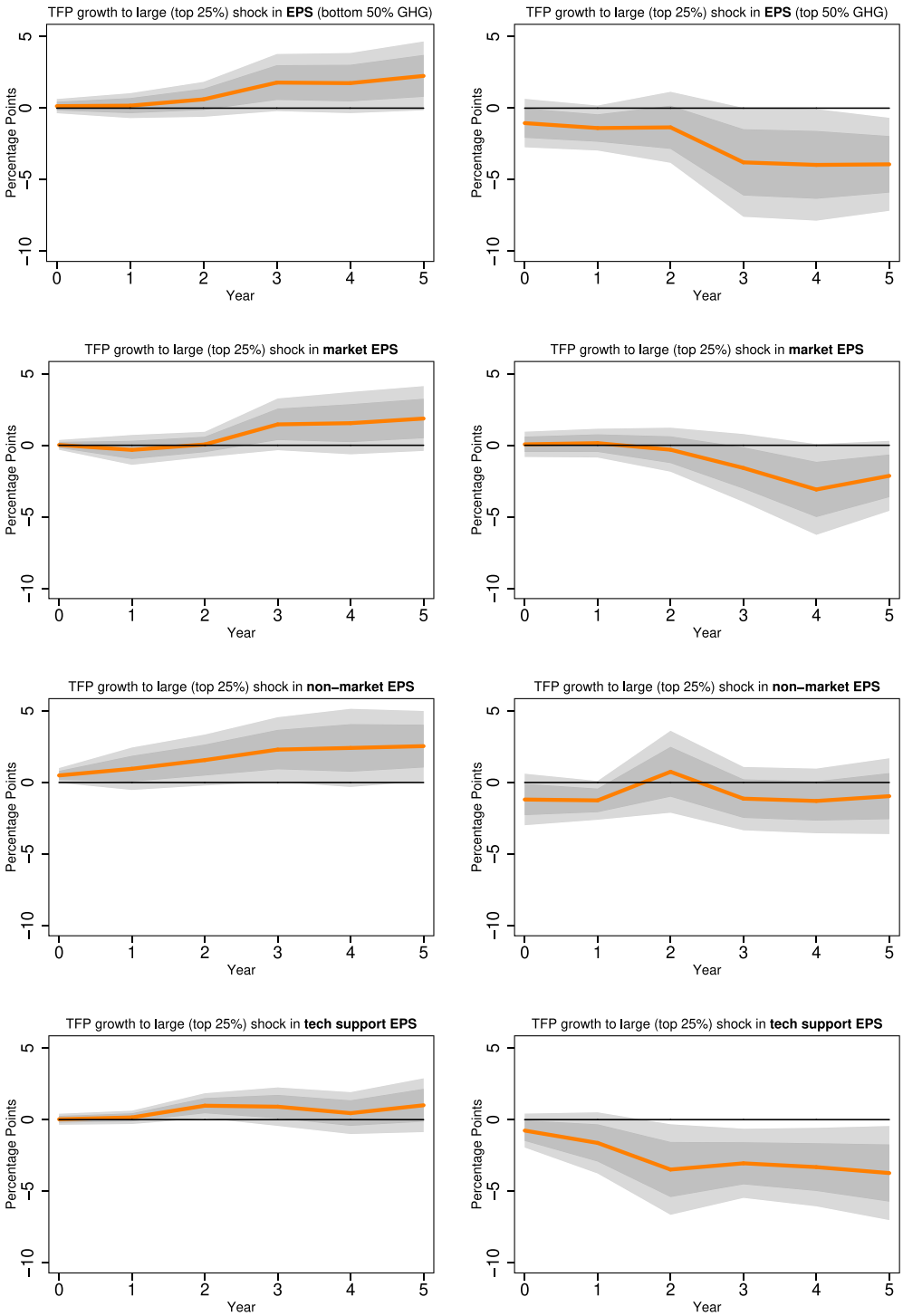

Fig. A.5. Impulse response functions – country-level – large shock.
Note: Cumulative impulse responses of labour productivity growth to 1 pp EPS shocks (top 25% changes) over 5 years. Left column: bottom 50% polluting countries, right column: top 50% polluting countries. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

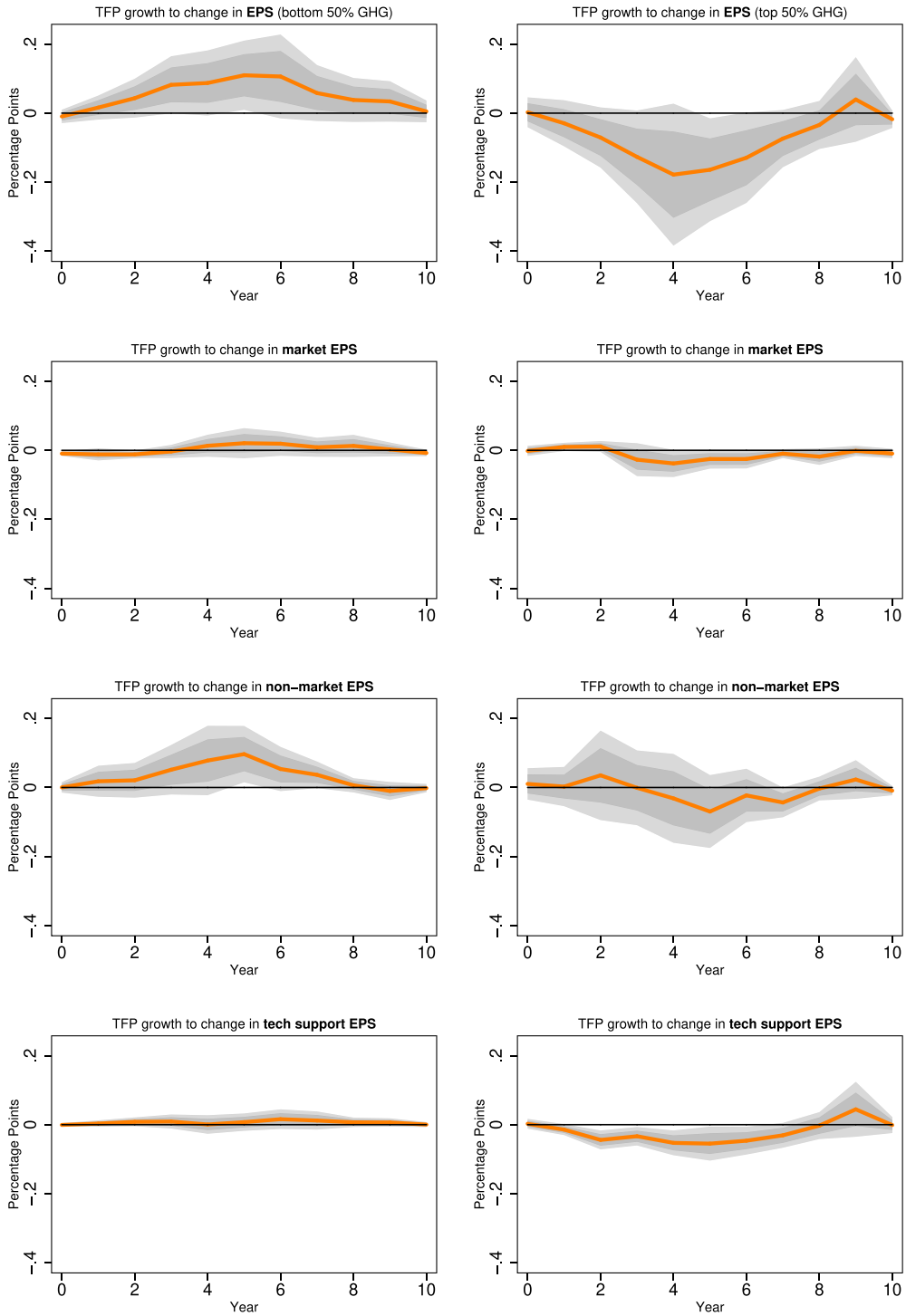

Fig. A.6. Impulse response functions – country level – all shocks.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive and negative changes) over 10 years. Left column: bottom 50% polluting countries, right column: top 50% polluting countries. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

Table A.4
Summary statistics — firm characteristics.

	Mean	SD	25th	Median	75th
Belgium					
TFP	1.16	0.81	0.73	0.97	1.34
Distance to TFP frontier	1.81	0.72	1.33	1.79	2.26
Labour prod	109,000.82	221,453.83	51,635.35	73,844.31	109,406.36
Distance to LP frontier	2.04	0.92	1.46	1.99	2.61
Employees	17.22	401.84	2.00	4.00	9.00
Firm age	21.55	13.49	12.00	19.00	27.00
ROA	14.63	15.51	6.08	12.56	21.21
Germany					
TFP	1.16	0.83	0.74	0.98	1.32
Distance to TFP frontier	1.88	0.69	1.40	1.85	2.33
Labour prod	116,644.23	219,176.73	58,899.91	83,144.93	119,460.07
Distance to LP frontier	2.17	0.86	1.61	2.15	2.71
Employees	58.52	908.50	4.00	13.00	35.00
Firm age	30.06	25.97	16.00	23.00	35.00
ROA	12.93	13.86	5.12	11.29	19.42
Spain					
TFP	1.32	1.49	0.64	0.90	1.35
Distance to TFP frontier	1.87	0.85	1.34	1.87	2.42
Labour prod	88,150.00	220,681.33	30,062.11	46,121.91	76,834.23
Distance to LP frontier	2.54	1.08	1.89	2.53	3.20
Employees	20.03	6,143.86	2.00	4.00	10.00
Firm age	16.32	9.81	9.00	15.00	21.00
ROA	20.65	75.48	1.85	6.70	14.64
France					
TFP	1.17	0.66	0.75	1.01	1.39
Distance to TFP frontier	1.77	0.71	1.27	1.72	2.22
Labour prod	72,004.61	91,730.05	38,492.93	56,565.88	82,407.70
Distance to LP frontier	2.31	0.89	1.71	2.23	2.85
Employees	29.40	599.24	2.00	5.00	12.00
Firm age	18.09	13.00	9.00	15.00	24.00
ROA	10.83	14.83	3.28	9.78	18.15
Italy					
TFP	1.78	3.00	0.56	0.84	1.43
Distance to TFP frontier	1.85	0.92	1.32	1.88	2.44
Labour prod	166,921.68	422,712.02	39,600.27	64,479.47	119,990.00
Distance to LP frontier	2.20	1.10	1.55	2.21	2.85
Employees	19.91	206.89	2.00	6.00	13.00
Firm age	20.95	13.09	12.00	18.00	28.00
ROA	33.20	83.89	4.15	8.67	17.89
Portugal					
TFP	1.24	0.89	0.73	1.05	1.50
Distance to TFP frontier	1.95	0.78	1.42	1.88	2.41
Labour prod	36,829.85	107,960.40	15,401.03	24,197.47	38,761.56
Distance to LP frontier	3.39	1.02	2.74	3.29	3.94
Employees	10.79	104.68	2.00	3.00	7.00
Firm age	20.28	14.14	11.00	17.00	25.00
ROA	6.52	23.79	0.96	6.67	15.05
All					
TFP	1.39	1.84	0.66	0.94	1.40
Distance to TFP frontier	1.85	0.83	1.33	1.84	2.37
Labour prod	100,967.75	267,763.09	31,020.16	51,867.29	86,211.59
Distance to LP frontier	2.47	1.10	1.78	2.43	3.13
Employees	23.68	3,438.03	2.00	5.00	12.00
Firm age	19.63	14.49	10.00	17.00	25.00
ROA	19.51	62.55	2.88	8.25	16.96

Table A.5
Variable description.

Variable	Country	Firm	Description	Source
Total factor productivity	X		TFP index as derived by Eurostat, 2015 = 100, growth rate	AMECO
		X	Based on ORBIS/iBACH data, Petrison–Levin–Wooldridge approach (Ackerberg et al., 2015), growth rate	own calc
TFP frontier	X		TFP growth rate of the US	AMECO
		X	average TFP growth of top 5% most productive firms per (2-digit NACE) sector and year	own calc
Distance to TFP frontier		X	Difference between 95th TFP percentile (per sector and year) and individual TFP level	own calc
Labour productivity	X		GDP per hour worked, constant prices 2010 and PPPs, growth rate	OECD
		X	Value added/number of employees from ORBIS/iBACH data, growth rate	own calc
Labour productivity frontier	X		GDP per hour worked of the US, growth rate	OECD
		X	average labour productivity growth of top 5% most productive firms per (2-digit NACE) sector and year	own calc
Distance to Labour prod frontier		X	Difference between 95th labour productivity percentile (per sector and year) and individual labour productivity level	own calc
Environmental policy stringency	X	X	EPS index, values between 0 and 6 (least to most stringent), aggregate or sub-indicator, growth rate	OECD
Emission indicator	X		Green house gas emissions per capita, dummy variable equals one if country is among top (50%) polluters	AMECO
		X	Dummy variable equals one if firm is among top X emission equivalence bins according to XGBoost classification, based on Urgentem data	own calc
GDP/capita	X	X	real Gross Domestic Product per capita	AMECO
Output gap	X	X	Cyclical component of real GDP	AMECO
Employment prot.	X	X	Strictness of employment protection legislation	OECD
Startup costs	X	X	Costs of starting a business for men as percentage of income per capita	World Bank
R & D spending	X		Research and Development investment as percentage of GDP, measured in constant prices of 2015 and PPPs	OECD
Industry share	X		Gross value added industry (NACE section C+D+E) as percentage of gross value added overall	AMECO
Firm size		X	Number of employees, standardized	ORBIS/iBACH
Firm age		X	Years since formation of business, standardized	ORBIS/iBACH
Firm leverage		X	Debt as percentage of assets, standardized	ORBIS/iBACH
Return-on-assets		X	Profit as percentage of assets, standardized	ORBIS/iBACH
Equity ratio		X	Equity per total assets, standardized	ORBIS/iBACH
Patents		X	Cumulative number of patent families, standardized	Orbis IP

Table A.6
Emission regressions.

	Top 6 emission bin — dummy				Emission bin (0-9)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔEPS	0.00916				−0.00002			
	(0.00682)				(0.00001)			
$\Delta market EPS$		0.00137				−0.000005**		
		(0.00193)				(0.000002)		
$\Delta non-market EPS$			0.00649				0.000007	
			(0.00398)				(0.000005)	
$\Delta techsupport EPS$				0.00128				−0.00001***
				(0.00241)				(0.000003)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm F.E.	No	No	No	No	Yes	Yes	Yes	Yes
R^2 adjusted					0.71	0.71	0.71	0.71
N	6,167,610	6,167,610	6,167,610	6,167,610	6,286,273	6,286,273	6,286,273	6,286,273

Note: Local projections 3 year ahead, (1)-(4) logit regression with top 6 emission bin dummy as dependent variable, (5)-(8) linear regression with emission bins as dependent variable. Controls include: output gap, gdp/capita, EPL, gov R&D expenditure, startup costs, TFP frontier growth, distance to frontier, firm size, age and ROA. * < 0.1, ** < 0.05, *** < 0.01

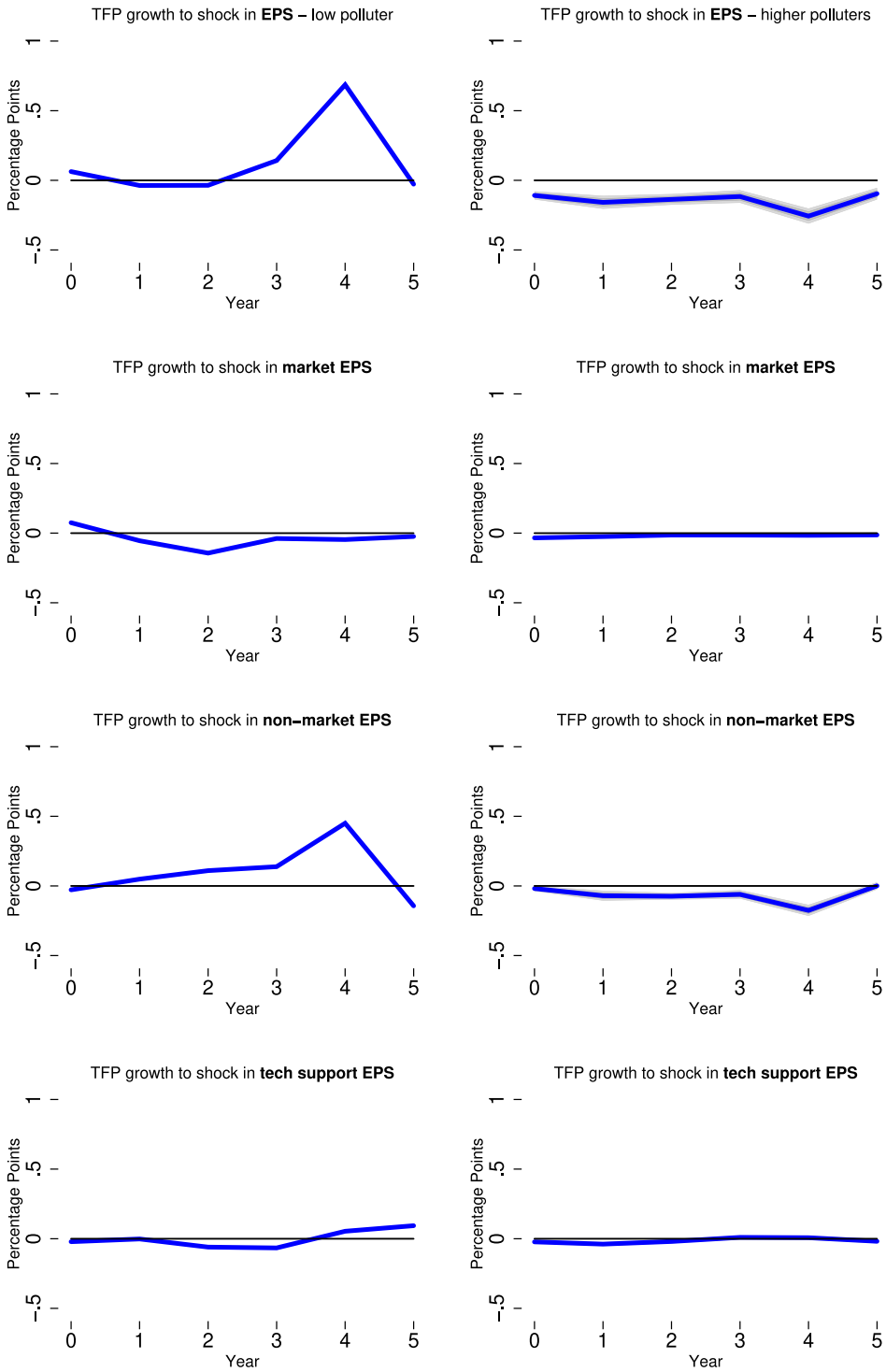

Fig. A.7. Impulse response functions – firm-level – 10 bins.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive changes) over 5 years. Left column represents baseline (= bottom emission bin), right column captures the linear interaction effect of higher bins of polluting firms. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

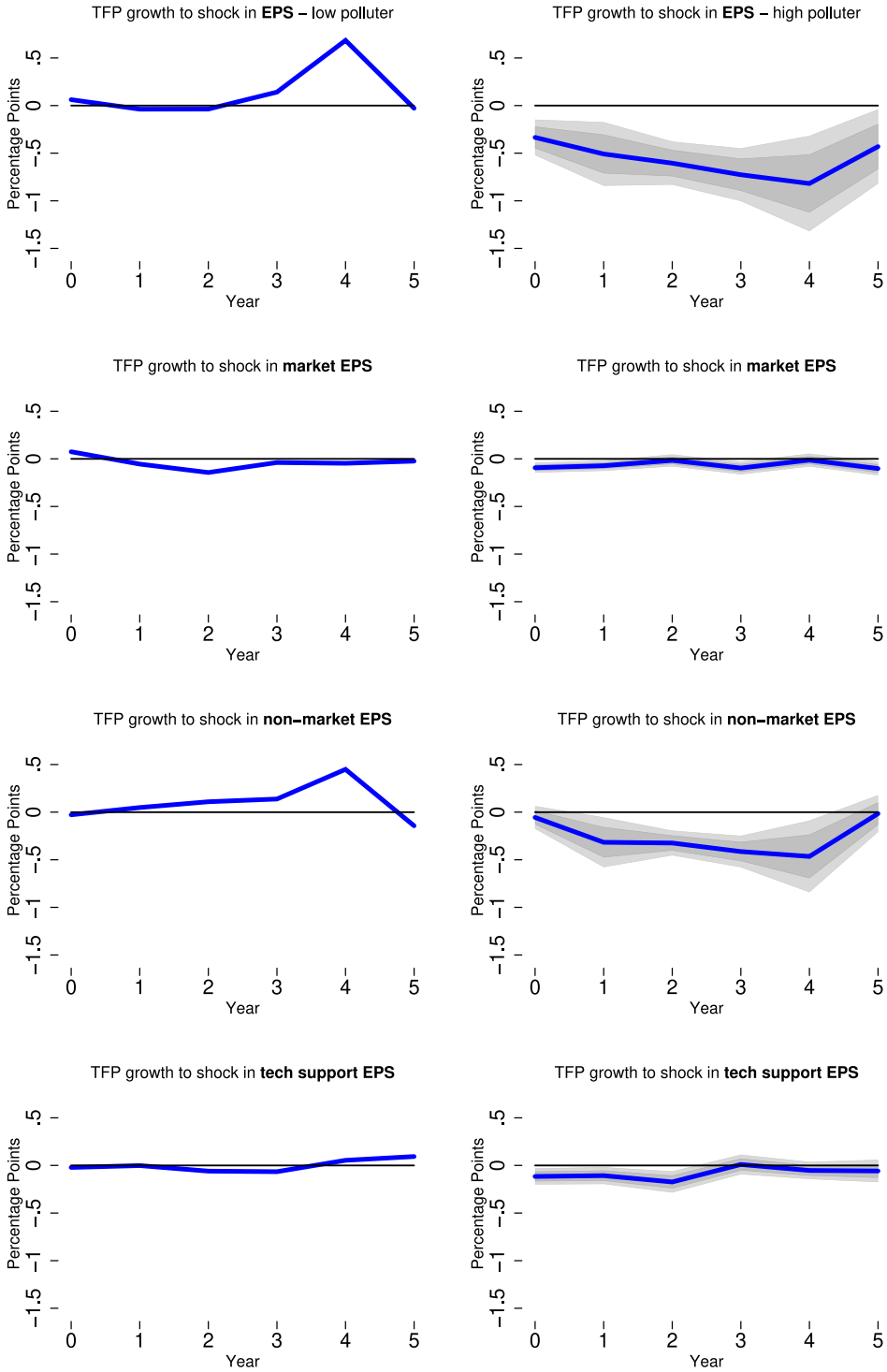

Fig. A.8. Impulse response functions – firm-level – initial bin.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive changes) over 5 years. Left column: bottom 4 bins of polluting firms, right column: top 6 bins of polluting firms (in first year). Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

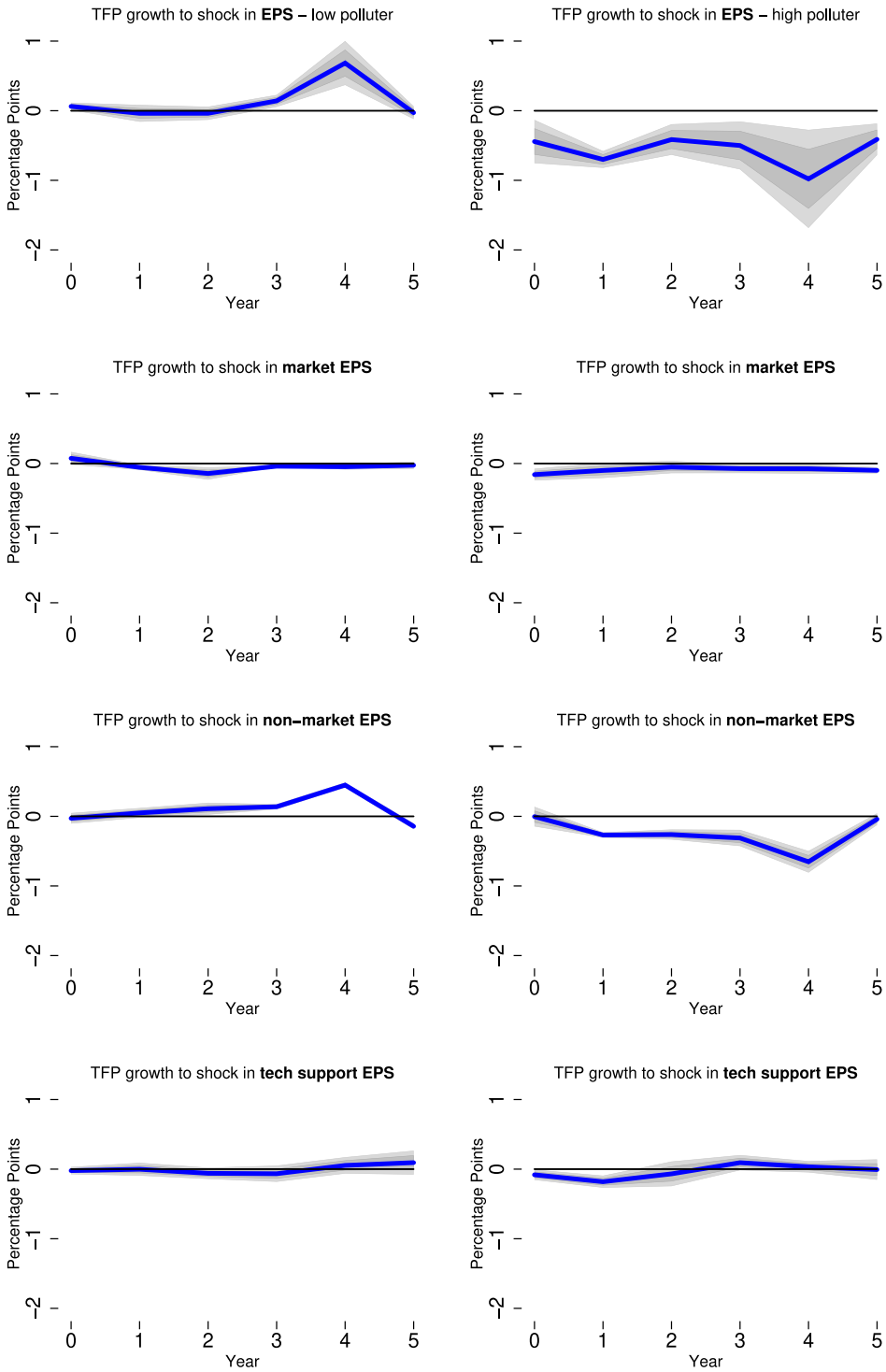

Fig. A.9. Impulse response functions – firm-level – country cluster SE.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive changes) over 5 years. Left column: bottom 4 bins of polluting firms, right column: top 6 bins of polluting firms. Blue line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands based on country-cluster standard errors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

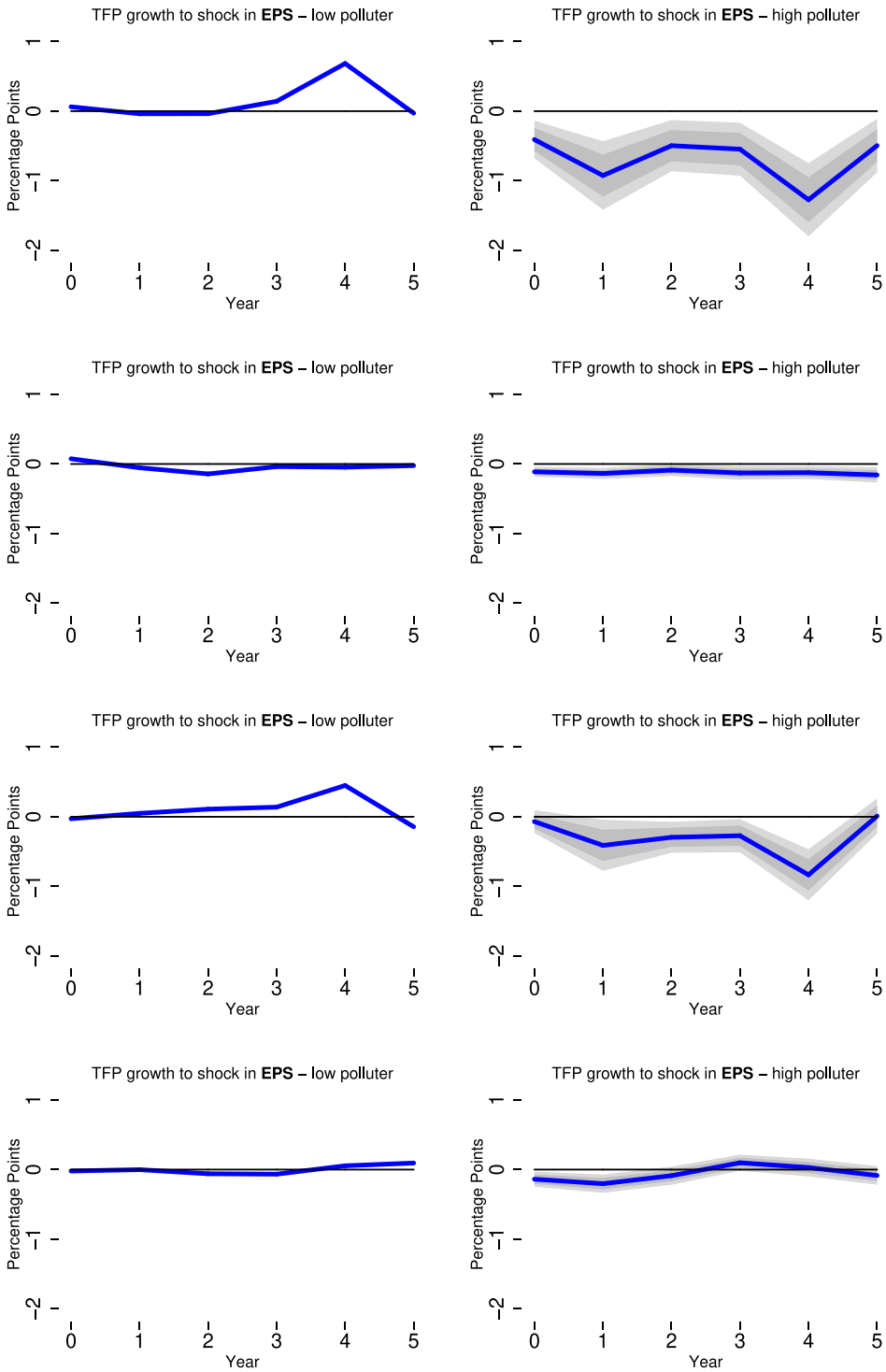

Fig. A.10. Impulse response functions – firm-level – top 4 bins.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive changes) over 5 years. Left column: bottom 6 bins of polluting firms, right column: top 4 bins of polluting firms. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

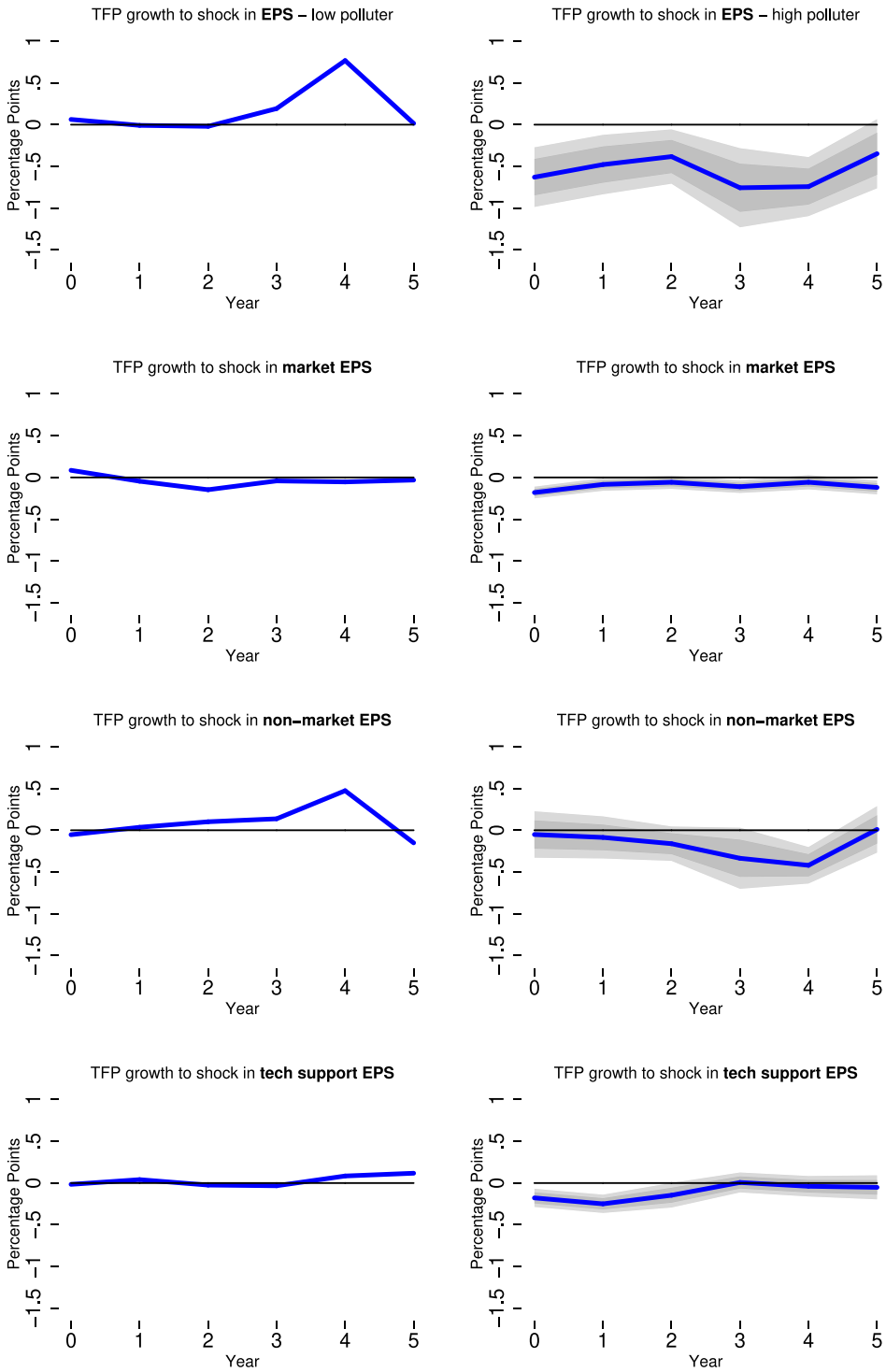

Fig. A.11. Impulse response functions – firm-level – labour productivity.
Note: Cumulative impulse responses of labour productivity growth to 1 pp EPS shocks (positive changes) over 5 years. Left column bottom 4 bins of polluting firms, right column top 6 bins of polluting firms. Blue line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

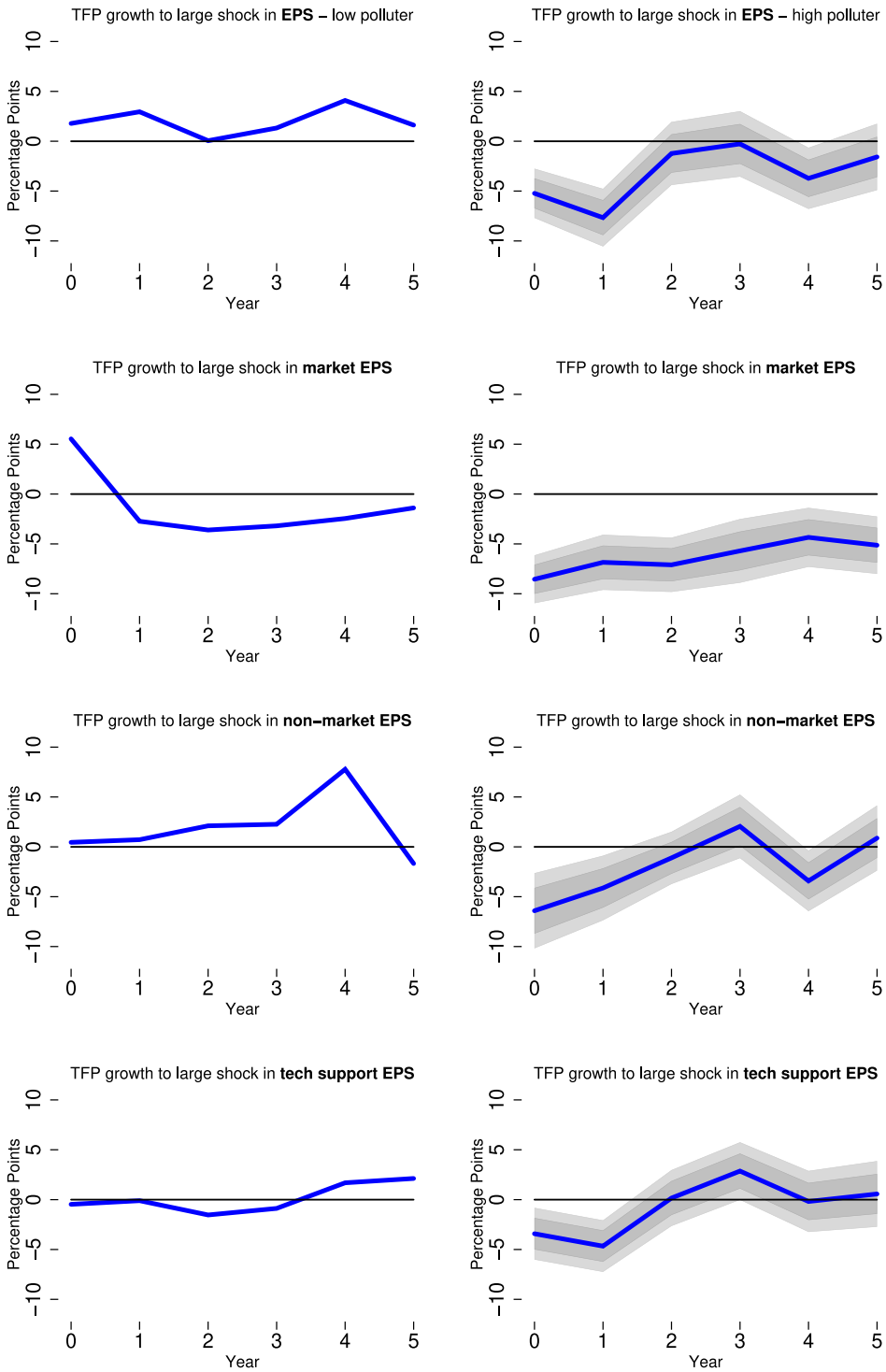

Fig. A.12. Impulse response functions – firm-level – large shocks.
Note: Cumulative impulse responses of TFP growth to a large EPS shock (top 25% changes) over 5 years. Left column: bottom 4 bins of polluting firms, right column: top 6 bins of polluting firms. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

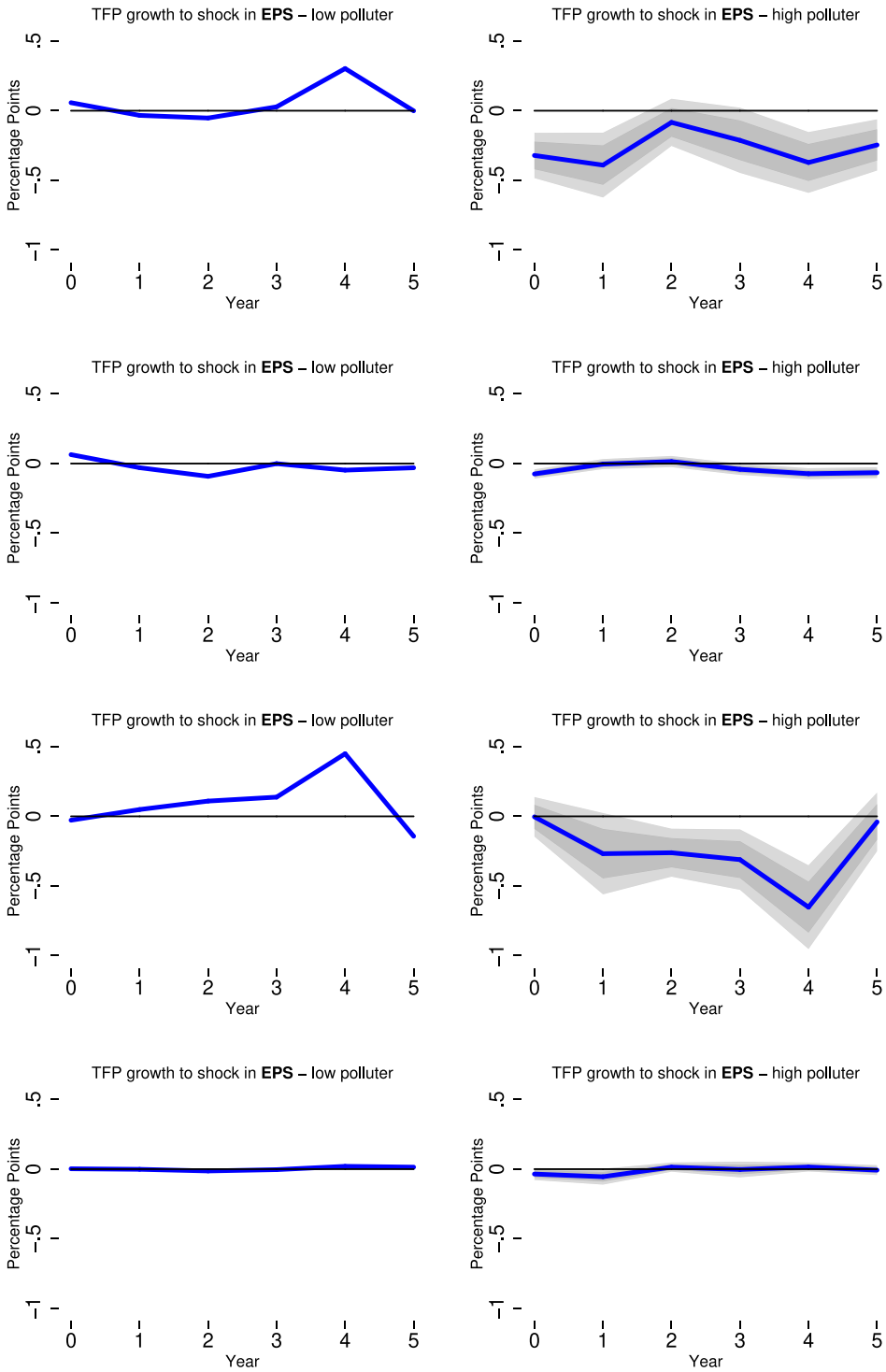

Fig. A.13. Impulse response functions – firm-level – all shocks.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive and negative changes) over 5 years. Left column bottom 4 bins of polluting firms, right column top 6 bins of polluting firms. Solid line represents mean responses, dark grey area 68% confidence bands, light grey area 90% confidence bands.

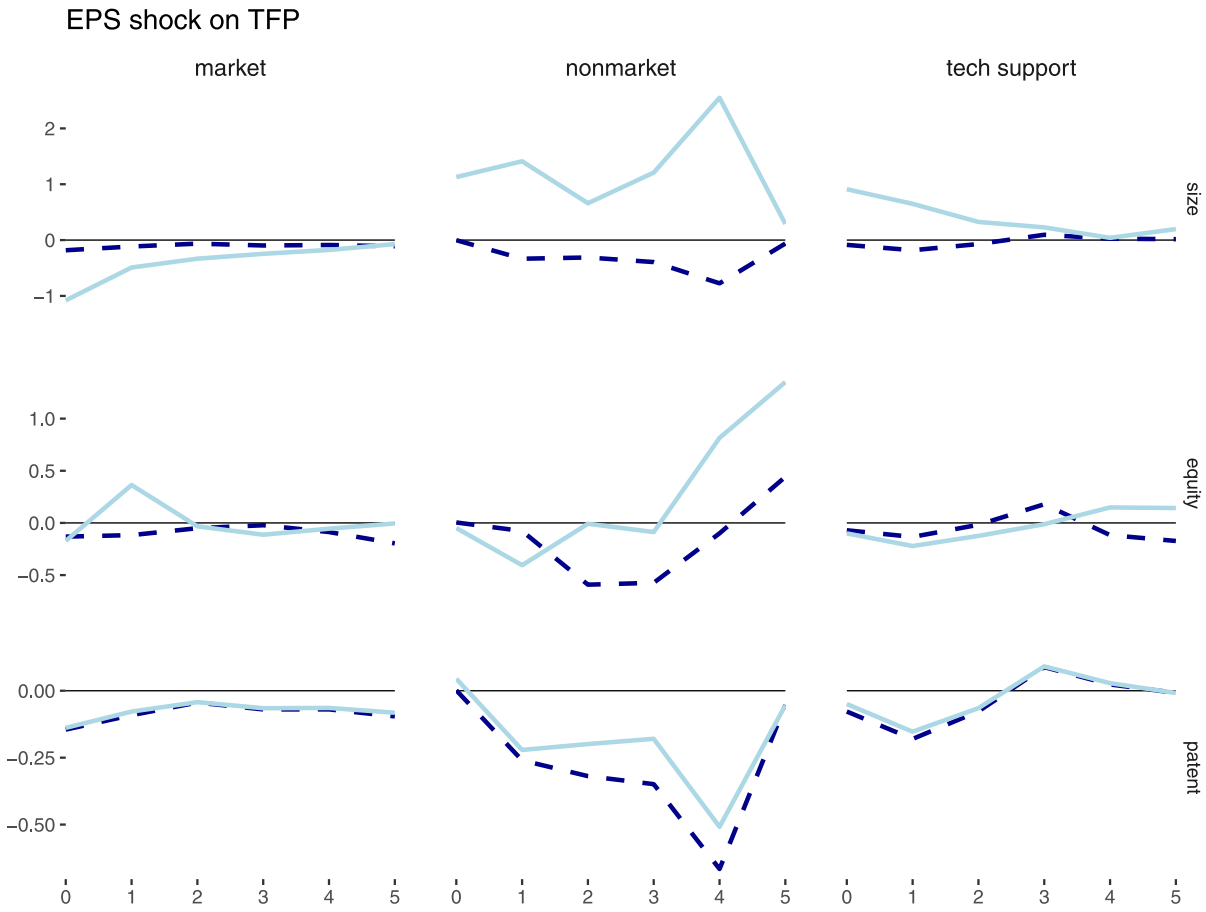

Fig. A.14. Impulse response functions – firm-level – heterogeneity.
Note: Cumulative impulse responses of TFP growth to 1 pp EPS shocks (positive changes) over 5 years for firms with different characteristics according to size, equity ratio and patenting experience. Light blue (solid) line — 90th percentile, dark blue (dashed) line — 10th percentile of the respective (size/equity/patenting) distribution.

Table A.7
Detailed local projection results.

	Country TFP growth		Firm TFP growth	
	(1y)	(5y)	(1y)	(5y)
$CO2_{i-1}$ indicator	−0.564	0.628	4.226***	3.921**
	(0.718)	(1.169)	(1.633)	(1.727)
EPS change	0.022	0.165	−0.035***	−0.029***
	(0.031)	(0.098)	(0.005)	(0.004)
$CO2_{i-1}$*EPS	−0.045	−0.230	−0.691***	−0.415**
	(0.055)	(0.135)	(0.220)	(0.198)
Output gap	0.469**	0.403	−1.604***	−2.541***
	(0.153)	(0.365)	(0.218)	(0.031)
GDP/capita	−24.49***	−81.551***	118.676***	135.249***
	(1.939)	(20.238)	(1.629)	(2.245)
EPL	8.029	17.823	−15.398***	−0.701
	(8.295)	(13.995)	(0.395)	(0.570)
Startup costs	−2.059**	−3.610*	−7.399***	−1.755***
	(0.942)	(1.732)	(0.108)	(0.113)
agg R&D	−2.094	4.398	−19.971***	8.569***
	(2.138)	(5.934)	(0.416)	(0.565)
agg TFP frontier	8.213***	−3.536		
	(1.239)	(12.052)		
sector TFP frontier			52.229***	57.053***
			(0.142)	(0.156)
Distance to frontier			81.700***	96.730***
			(0.075)	(0.089)
Size (employees)			−0.620***	−0.098
			(0.171)	(0.096)
Return-on-Assets			−4.347***	−4.093***
			(0.051)	(0.058)
Age			9.807***	11.828***
			(0.490)	(0.807)
Country F.E.	Yes	Yes	No	No
Time F.E.	Yes	Yes	Yes	Yes
Firm, sector F.E.	No	No	Yes	Yes
R^2 adjusted	0.04	0.01	0.43	0.54
N	182	134	8,206,621	4,422,580

Note: Local projections with 1 and 5 year ahead cumulative TFP growth as dependent variable at country and firm level. CO2 indicator is top 50% of countries according to emission intensity or top 6 bins of polluting firms. All controls are lagged except for output gap. Age, size and ROA are standardized. * < 0.1, ** < 0.05, *** < 0.01

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